

**AN EXPLAINABLE FEATURE-BASED SMARTPHONE
RECOMMENDATION SYSTEM USING ASPECT-BASED
SENTIMENT ANALYSIS (ABSA) OF USER REVIEWS**

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Declaration

I declare that this dissertation is my own work and has not been submitted, either in whole or in part, for any other degree or academic qualification. All sources used in developing the study have been acknowledged through appropriate citations and references.

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Date: ____20/09/2026____

Abstract

Smartphone purchasers must compare many products and interpret large volumes of online reviews before making a decision. Aggregate ratings and behaviour-based recommenders can obscure feature-level strengths and weaknesses and may provide limited reasons for a recommendation. This study developed an explainable feature-based smartphone recommendation system that transformed Amazon smartphone reviews into structured evidence through Aspect-Based Sentiment Analysis (ABSA).

A pragmatic, predominantly quantitative design-and-development approach was used. A frozen Bright Data Amazon review export was screened and stored in SQLite. The final corpus contained 4,079 reviews for 223 eligible smartphones; preprocessing retained 3,388 English reviews, which were segmented into 31,683 sentences. A transparent lexicon-first ABSA engine with contextual polarity rules identified battery, camera, display, performance, design, and price, producing 13,914 aspect-sentiment records. Aggregation and Bayesian shrinkage produced 1,240 phone-aspect score rows and at least one feature score for 221 smartphones. User priorities were normalized and combined with the feature scores to generate Top-N rankings and contribution-based explanations.

Evaluation showed that text polarity agreed directionally with Amazon star ratings for approximately 67% of 2,925 comparable reviews; agreement was stronger for four- and five-star reviews (approximately 70–79%) than for one- to three-star reviews (approximately 33–37%). This was treated as a sanity check rather than classification accuracy because aspect opinions and overall ratings measure different constructs. Twelve feedback sessions produced 90 phone ratings, a mean satisfaction score of approximately 3.49 out of 5, and NDCG@3 of approximately 0.56; about 56% of ratings were four or five. These results indicate mixed satisfaction and weak-to-moderate ranking quality. The Amazon-star baseline was implemented, but no paired baseline-rated runs were available, so superiority was not claimed. No gold-labelled Macro-F1 result was reported. The completed system demonstrates a feasible and auditable route from review evidence to preference-weighted recommendation, while showing that ranking quality and comparative evaluation require further improvement.

Keywords: Aspect-Based Sentiment Analysis; smartphone recommendation; explainable artificial intelligence; Amazon reviews; feature-based recommendation; sentiment-aware recommendation.

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Acronyms

Acronym	Meaning
ABSA	Aspect-Based Sentiment Analysis
AI	Artificial Intelligence
API	Application Programming Interface
ASIN	Amazon Standard Identification Number
BERT	Bidirectional Encoder Representations from Transformers
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
NDCG	Normalized Discounted Cumulative Gain
NLP	Natural Language Processing
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
TF-IDF	Term Frequency-Inverse Document Frequency
XAI	Explainable Artificial Intelligence

Chapter 1: Introduction

1.1 Chapter Introduction

The growth of e-commerce has expanded smartphone choice, but it has also increased the effort needed to compare models and interpret large volumes of reviews. Buyers usually evaluate several attributes-camera, battery, display, performance, design, and price while conventional systems often reduce these experiences to overall ratings or interaction patterns. Such outputs may identify popular products without clearly showing whether they match an individual user's priorities.

This study developed an explainable feature-based smartphone recommendation system using Aspect-Based Sentiment Analysis (ABSA) of Amazon user reviews. The system extracted feature-level sentiment, created product feature scores, applied user-defined weights, ranked smartphones, and explained the results in plain language. This chapter presents the research background, problem, aim, approved objectives and questions, significance, scope, completed research outcomes, and dissertation structure.

1.2 Background of the Study

1.2.1 Recommendation Systems in E-commerce

E-commerce recommendation systems commonly use collaborative filtering, content-based filtering, or hybrid models. These approaches can reduce search effort and model complex user-item relationships, but collaborative methods depend on sufficient historical interactions and may perform poorly for new users or products (Sami et al., 2024; Volkovs et al., 2017; Lin, 2022).

A second limitation is weak transparency. Smartphone buyers make different feature trade-offs, yet an overall score rarely shows whether a recommendation is driven by camera, battery, price, or another attribute. Aspect- and review-aware studies indicate that review sentiment can be represented as separate recommendation criteria, providing a clearer basis for ranking (Musto et al., 2017; Dang et al., 2021; Liu et al., 2021; Haghighi & Moradian Zadeh, 2025).

1.2.2 Smartphone Selection as a Multi-Feature Decision

Smartphone selection is a multi-criteria decision involving technical specifications and real-use experience. Specifications describe components such as processors, memory,

battery capacity, and camera resolution, but they do not fully reflect battery endurance, low-light image quality, heating, responsiveness, build quality, or value in daily use. Because users assign different importance to camera, battery, display, performance, design, and price, the same phone may be appropriate for one buyer and unsuitable for another. Treating these features as separate dimensions makes trade-offs visible and allows the final recommendation to be traced to feature scores and explicit user priorities (Musto et al., 2017; Dietz et al., 2022).

1.2.3 Role of Online User Reviews

Online reviews provide practical evidence about product use that is often absent from manufacturer descriptions. Smartphone reviewers discuss battery drain, camera quality, display brightness, gaming performance, heating, design, and value for money. However, the text is unstructured and may contain spelling errors, informal language, duplicate content, and several opinions in one sentence.

Research shows that mobile-phone and e-commerce reviews can be collected and analyzed at aspect level, with transformer models performing strongly in contextual review language (Soni et al., 2024; Singhal et al., 2025; Rasappan et al., 2024). The study therefore uses a cleaned offline Amazon dataset and applies duplicate, relevance, and traceability controls because manipulated or repeated reviews can distort product scores (Jin et al., 2026).

1.2.4 Aspect-Based Sentiment Analysis

ABSA identifies the feature discussed in a text and the sentiment directed toward it. Unlike document-level classification, it can preserve different opinions within the same review; for example, ‘the camera is excellent, but the battery drains quickly’ produces camera-positive and battery-negative records.

Recent ABSA research combines contextual transformer representations with linguistic, graph, sequential, or extraction components to improve aspect detection and sentiment classification (Aziz et al., 2024; Jayakody et al., 2024). Mobile-phone evidence also supports fine-tuned BERT models for technical review language (Singhal et al., 2025). These studies informed the literature review; however, the implemented system used a transparent, versioned lexicon-first engine with contextual polarity rules and a predefined six-aspect scheme. An external large-language-model interface remained optional and was not the source of the reported labels.

1.2.5 Feature Scoring and Explainable Recommendation

The classified aspect records are aggregated into separate product scores for camera, battery, display, performance, design, and price. Users assign importance weights to these features, and the recommendation engine combines the weighted values to rank eligible smartphones. This approach supports personalization without requiring a previous interaction history (Musto et al., 2017; Haghighi & Moradian Zadeh, 2025). Explanation is generated from the same evidence used for ranking. The interface can show which highly weighted features increased a phone's score and identify important trade-offs. This links analytical outputs to understandable reasons and avoids presenting an unexplained recommendation (Mansilla et al., 2025; Chen et al., 2026; Zhu, 2025).

1.3 Research Problem

Current smartphone recommendation methods commonly rely on overall ratings, historical interactions, or similarity patterns. Although useful for general ranking, they may conceal feature-level strengths and weaknesses. A highly rated phone can still receive repeated criticism for battery or camera performance, while a moderately rated device may perform strongly in the feature a particular buyer values most.

The reviewed literature confirms that ABSA can extract feature sentiment and that review-derived sentiment can support recommendation (Soni et al., 2024; Dang et al., 2021; Liu et al., 2021). Nevertheless, ABSA studies often end with classification metrics, while recommender studies frequently emphasize rating prediction or latent user-item modelling. Limited work integrates smartphone review collection, explicit feature scoring, user-defined weights, transparent ranking, and plain-language explanations in one system.

The research problem is therefore the insufficient transformation of smartphone review text into feature-level, user-priority-based, and explainable recommendations. This study addresses that problem through an end-to-end Amazon review, ABSA, scoring, ranking, and explanation framework.

1.4 Research Aim

The aim of this research is:

To develop an explainable feature-based smartphone recommendation system using Aspect-Based Sentiment Analysis (ABSA) of user reviews.

The system analysed textual reviews, extracted feature-level sentiments, computed evidence-aware feature performance scores, and generated transparent, requirement-driven recommendations without relying on user history.

1.5 Research Objectives

The following goals are established in order to fulfill the research aim:

1. To collect and prepare a structured dataset of smartphone reviews from Amazon for natural language processing and analysis
2. To identify predefined smartphone features from review text and classify the sentiment associated with each feature as positive, negative, or neutral using Aspect-Based Sentiment Analysis (ABSA).
3. To determine each smartphone's feature performance score by calculating the percentage of positive sentiment mentions for each feature.
4. To develop a recommendation engine that ranks smartphones using weighted feature scores based on user-defined feature preferences.
5. To produce AI explanation messages in plain language that support each suggestion using the actual feature scores.

1.6 Research Questions

This study seeks to answer the following research questions:

- 1. How can sentiment at the feature level be reliably extracted from reviews of smartphones?**
- 2. How can sentiment scores for features be combined to create useful performance metrics?**
- 3. How can explainable recommendations be produced using feature scores?**
- 4. When compared to conventional rating-based systems, does feature-based recommendation enhance interpretability**

Table 1.1: Alignment of Research Questions and Research Objectives

Research Question	Related Objective(s)	Purpose
RQ1: Feature-level sentiment extraction	Objectives 1 and 2	To prepare the review dataset, identify the predefined smartphone features, and classify sentiment for each detected feature.
RQ2: Feature performance measurement	Objective 3	To convert classified feature sentiments into measurable performance scores for each smartphone.
RQ3: Explainable recommendation generation	Objectives 4 and 5	To rank smartphones according to user-defined priorities and explain the result using actual feature scores.
RQ4: Interpretability compared with rating-based recommendation	Objectives 4 and 5; system evaluation	To compare the clarity and usefulness of the feature-based output with a conventional rating-based baseline without introducing an additional objective.

The fourth research question is evaluated through the system-evaluation phase described in Chapter 3. It does not create an additional research objective because the approved proposal contains five objectives. The comparison examines whether recommendations produced through Objectives 4 and 5 are more understandable than a basic rating-based ranking.

1.7 Significance of the Research

The study contributes to consumer decision support, recommendation-system design, applied ABSA, and explainable artificial intelligence by linking feature-level review evidence directly to transparent smartphone ranking.

1.7.1 Significance for Smartphone Buyers

Buyers can compare feature profiles instead of reading large numbers of reviews manually. Because the ranking reflects explicit priorities, photography-focused, battery-focused, performance-focused, and price-focused users can receive different results from the same product dataset. Visible feature contributions and trade-offs may also improve decision confidence (Dietz et al., 2022).

1.7.2 Significance for Recommendation-System Research

The study offers an auditable alternative to methods based mainly on ratings, latent profiles, or interaction history. Review-aware research shows the value of aspect sentiment, while the proposed design makes the relationship between user preferences and product ranking directly visible through feature weights (Liu et al., 2021; Haghighi & Moradian Zadeh, 2025; Danish et al., 2026). It reduces new-user cold-start dependence, although products without sufficient reviews remain outside the system’s reliable scope.

1.7.3 Significance for ABSA and Natural Language Processing

The research demonstrates how ABSA outputs can be used beyond classification. Feature detection and sentiment labels were converted into product scores, recommendations, and explanations using real smartphone review language. Technical quality was assessed through corpus coverage, linked evidence, star–text directional agreement, ranking metrics, and user feedback. Because an independently gold-labelled benchmark was not completed, classification accuracy, precision, recall, and Macro-F1 were not reported.

1.7.4 Significance for Explainable Artificial Intelligence

The explanation module is grounded in the feature scores and user weights that determine the ranking. It therefore provides user-centered reasons-why a phone fits the

buyer’s priorities rather than only model-centered information about influential tokens or internal predictions (Mansilla et al., 2025; Chen et al., 2026).

1.7.5 Practical and Broader Value

Aggregated feature sentiment can also help manufacturers and e-commerce analysts identify commonly praised or criticized product attributes. The same pipeline could later be adapted to other multi-feature products by changing the aspect categories and domain vocabulary.

1.8 Scope and Delimitations of the Research

The study is limited to the design and evaluation of a research prototype for explainable smartphone recommendation. The boundaries are defined to keep the implementation feasible and to ensure that the system can be evaluated using a stable and traceable dataset.

1.8.1 Data Source and Collection Boundary

The main data source was a Bright Data export of publicly visible Amazon smartphone review text. Reviews were collected before system development and evaluation and stored as a frozen offline dataset. The system did not scrape Amazon in real time when a user requested a recommendation. This boundary supported reproducibility and prevented changes in live pages from affecting the reported results.

Review text is treated as secondary user-generated data. Private profiles, login-protected information, payment-restricted content, and personal purchasing histories are outside the scope. Reviewer names and identifiers are not required for the analysis.

1.8.2 Product and Language Boundary

The product category is limited to smartphones. Accessories, tablets, laptops, smartwatches, and other electronics are excluded. Restricting the study to one product category allows the aspect vocabulary, feature scoring, and evaluation procedures to remain consistent.

Only English-language reviews were used in the main implementation. Multilingual ABSA research demonstrates that cross-language review analysis is possible, but it also introduces challenges related to language imbalance, weak supervision, translation, and

domain transfer (Satria et al., 2025; Ratniasih et al., 2025). A multilingual extension was therefore reserved for future work.

1.8.3 Feature Boundary

The ABSA process is limited to six predefined features: camera, battery, display, performance, design, and price. These features form the feature-scoring and weighting structure approved in the research proposal. Other aspects such as speakers, biometric sensors, software updates, charging speed, network reception, and after-sales service are not included in the primary version.

1.8.4 Recommendation Boundary

The recommendation engine used review-derived feature scores and explicit user-defined feature weights. It did not use collaborative filtering, user similarity, long-term behaviour tracking, or purchase-history profiling. Overall Amazon star ratings were retained as supplementary information and as the implemented comparison baseline, but they did not determine the proposed feature-based ranking.

The system addresses user cold-start by allowing a new user to state feature priorities. It does not claim to solve the new-item cold-start problem for products that have no reviews, because feature performance scores cannot be calculated without sufficient review evidence.

1.8.5 Evaluation Boundary

Evaluation examined the end-to-end data funnel, feature and sentiment coverage, traceability of explanations, star-text directional agreement, user ratings, and Top-N ranking quality. Twelve feedback sessions yielded 90 phone ratings, a mean satisfaction score of approximately 3.49 out of 5, and NDCG@3 of approximately 0.56. Star-text agreement was approximately 67% for 2,925 comparable reviews and was used only as a broad sanity check. The Amazon-star baseline was implemented, but paired baseline-rated runs were not completed. Because no independently gold-labelled test set was available, accuracy, precision, recall, Macro-F1, and a confusion matrix were not claimed. The study did not predict future sales, commercial demand, or smartphone market prices.

Table 1.2: Scope and Delimitations of the Study

Area	Included in the Study	Not Included in the Study
Data source	Publicly visible Amazon smartphone review text collected into an offline dataset.	Private user data, login-protected content, and real-time scraping during recommendation.
Language	English-language reviews.	Multilingual and cross-language model evaluation.
Product category	Smartphones.	Accessories, tablets, laptops, smartwatches, and other electronics.
Features	Camera, battery, display, performance, design, and price.	Additional device-specific aspects outside the approved framework.
Recommendation logic	ABSA-derived feature scores and user-defined feature weights.	Collaborative filtering, user-similarity modelling, and long-term behavioral profiling.
Evaluation	ABSA metrics, ranking relevance, explanation clarity, and rating-based baseline comparison.	Sales forecasting, market-demand prediction, and production-level commercial deployment.

1.9 Research Outcomes

The completed research produced a frozen and traceable dataset of 4,079 reviews for 223 eligible smartphones, together with explicit preprocessing decisions that retained 3,388 usable English reviews.

The lexicon-first ABSA pipeline converted 31,683 sentences into 13,914 aspect-sentiment records for six smartphone features. Aggregation created 1,240 evidence-aware phone-aspect score rows and scoring coverage for 221 smartphones.

The FastAPI prototype converted normalized user priorities into Top-N recommendations, displayed contribution-based reasons, retained Amazon stars as a baseline, and exposed the same evidence through Recommend, Phones, Evidence, and Evaluation interfaces.

Evaluation provided an initial rather than definitive assessment: star-text agreement was approximately 67%, mean user satisfaction was approximately 3.49 out of 5, and NDCG@3 was approximately 0.56. The paired Amazon-star baseline evaluation and a gold-labelled ABSA benchmark remained incomplete, so the findings support feasibility and transparency rather than superiority or validated classification accuracy.

1.10 Structure of the Dissertation

The dissertation contains seven chapters. Chapter 1 introduces the study, its aim, objectives, questions, scope, and principal outcomes. Chapter 2 reviews the selected

literature, reports the systematic-review process, identifies the research gap, and presents the conceptual framework. Chapter 3 explains the pragmatic, predominantly quantitative design-and-development methodology, including data collection, lexicon-first ABSA, scoring, ranking, explanation, evaluation, quality assurance, ethics, and the four interface modules. Chapter 4 reports the corpus, ABSA, feature-score, star-text, and user-evaluation results. Chapter 5 discusses those findings in relation to the research questions and literature. Chapter 6 presents the conclusions and recommendations, and Chapter 7 states the limitations and future work. The reference list follows the final chapter.

1.11 Chapter Summary

This chapter established smartphone selection as a multi-feature decision for which overall ratings and interaction-based recommendations may provide insufficient detail. It introduced Amazon reviews as experiential evidence and ABSA as the method used to separate opinions about camera, battery, display, performance, design, and price.

The chapter also stated the research problem, aim, five approved objectives, four research questions, significance, scope, and completed research outcomes. The results showed that the implemented pipeline was feasible and explainable, while the initial user evaluation indicated mixed satisfaction and weak-to-moderate Top-N quality. Chapter 2 evaluates the selected literature and develops the conceptual framework for the implemented system.

Chapter 2: Literature Review

2.1 Chapter Introduction

This chapter reviews research supporting an explainable feature-based smartphone recommendation system using ABSA. The central problem is that users receive many ratings and reviews but limited evidence showing whether a phone fits the features they value. Conventional systems can perform well when interaction data are available, yet they may retain cold-start limitations, hide feature trade-offs, and provide weak explanations (Musto et al., 2017; Dang et al., 2021; Liu et al., 2021; Danish et al., 2026). The review therefore focuses on recommendation systems, online reviews, ABSA methods, sentiment-aware recommendation, explainability, and smartphone or mobile-review analysis. It also reports the systematic-review process for the finalized 32-paper corpus, compares the major methods, identifies the research gap, and presents the conceptual framework.

2.2 Product Recommendation Systems

2.2.1 Traditional Recommendation Approaches

Collaborative filtering recommends items from user-item interaction patterns, content-based filtering matches item characteristics to user profiles, and hybrid systems combine several signals. These approaches reduce information overload, but they may not reveal which smartphone features explain a ranking and can be limited by sparse or unavailable interaction histories (Musto et al., 2017; Sami et al., 2024; Danish et al., 2026).

2.2.2 Limitations of Rating- and Interaction-Based Recommendation

Cold-start occurs when new users or products provide insufficient interaction data. Content and graph-based models can reduce this limitation by adding item, review, or network information, although they may still require interaction structures (Volkovs et al., 2017; Lin, 2022).

A further problem is the loss of feature-level detail. Overall ratings may hide repeated complaints about battery, heating, or camera quality, and ranked outputs often provide little explanation. For high-involvement purchases, users need visible reasons and trade-

offs rather than only a predicted score (Dietz et al., 2022; Chen et al., 2026; Gajula, 2025).

2.2.3 Movement toward Review-Aware and Sentiment-Aware Recommendation

Review-aware systems use textual opinions to enrich user and product representations. Aspect sentiment has improved multi-criteria, deep, multilingual, and hybrid recommendation models (Musto et al., 2017; Dang et al., 2021; Liu et al., 2021; Danish et al., 2026). However, much of this work emphasizes rating prediction or latent profiles, whereas the present study uses explicit feature scores and user-defined weights for transparent smartphone ranking.

2.3 Online User Reviews and Smartphone Product Evaluation

2.3.1 Online Reviews as Product Experience Data

Online reviews describe real or perceived product experience, including battery endurance, camera performance, display quality, heating, responsiveness, design, and value. These signals complement specifications and provide richer evidence than star ratings alone. Review-derived aspect and sentiment information has therefore become an important input to recommendation research (Musto et al., 2017; Dang et al., 2021; Haghighi & Moradian Zadeh, 2025).

2.3.2 Importance of Feature-Level Analysis in Smartphone Recommendation

Smartphones combine strengths and weaknesses across several attributes, so the same product can suit one user and disappoint another. ABSA preserves these differences by assigning sentiment separately to each feature; for example, display-positive and battery-negative can coexist within one sentence (Soni et al., 2024; Singhal et al., 2025; Kumari et al., 2025).

2.3.3 Review Credibility and Data Quality

Review-based scoring is sensitive to duplicate, promotional, spam, and low-information content. Spam-aware research shows that misleading reviews can affect rankings and trust (Jin et al., 2026). The proposed system therefore includes deduplication, language

and relevance screening, traceable source records, and minimum evidence requirements.

2.4 Aspect-Based Sentiment Analysis

2.4.1 Concept and Core Tasks of ABSA

ABSA identifies product aspects and the sentiment expressed toward each one. Its tasks range from aspect and opinion extraction to aspect-sentiment pairing, triplet extraction, and aspect-category sentiment classification (Aziz et al., 2024; Maldonado Castillo et al., 2024). In this study, the output is structured feature sentiment that can be aggregated for recommendation.

2.4.2 Lexicon-Based and Classical Machine Learning Approaches

Classical ABSA uses lexicons, part-of-speech information, dependency rules, feature selection, and classifiers such as Naïve Bayes. These methods are efficient and interpretable but depend on feature engineering and may struggle with implicit or context-dependent sentiment (Mubarak et al., 2017). Recent syntactic approaches remain useful for transparent aspect-opinion mapping, although contextual models generally handle noisy reviews more effectively (Mansilla et al., 2025).

2.4.3 Neural Network and Hybrid Deep Learning Approaches

CNN, recurrent, and BiLSTM models learn local and sequential patterns from review text, while hybrid models combine contextual embeddings with sequence or task-specific components. Studies using CNN and Word2Vec, hybrid e-commerce classifiers, BERT-BiLSTM, multimodal text-emoji features, and multitask learning demonstrate improved representation but also highlight data, tuning, and class-imbalance requirements (Lestari et al., 2024; Rasappan et al., 2024; Setiabudi & Faizah, 2025; Padigapati & Krishna, 2025; Chong et al., 2026).

2.4.4 Transformer-Based ABSA Models

Transformers such as BERT, DeBERTa, and XLM-RoBERTa model words in context, which is valuable when smartphone terms change meaning across features or sentences contain several opinions. Unified BERT-graph models and Instruct-DeBERTa illustrate

how contextual representations can support extraction and classification together (Aziz et al., 2024; Jayakody et al., 2024).

Mobile-phone studies also report stronger performance for BERT than several classical and neural baselines (Soni et al., 2024; Singhal et al., 2025). Transformer-based ABSA is therefore the strongest fit for the core classifier, while transparent aspect categories and validation remain necessary.

2.4.5 Multi-Domain and Multilingual ABSA

ABSA performance can decline across products, platforms, and languages. Multilingual studies using XLM-RoBERTa and contextual models show the potential for cross-language analysis but also expose imbalance, supervision, and transfer challenges (Satria et al., 2025; Ratniasih et al., 2025; Di Nuovo et al., 2024). The present system is therefore limited to English Amazon smartphone reviews, with multilingual extension reserved for future work.

2.5 Integration of ABSA with Recommendation Systems

2.5.1 Aspect Sentiment as Recommendation Evidence

Aspect sentiment can function as structured recommendation evidence. Multi-criteria and review-aware studies show that it can represent product strengths and user preferences more meaningfully than one overall rating (Musto et al., 2017; Liu et al., 2021; Haghighi & Moradian Zadeh, 2025). The proposed system makes this evidence visible through product feature scores.

2.5.2 Multi-Criteria and Feature-Based Recommendation

Each smartphone feature can be treated as an independent recommendation criterion. User-defined weights then determine how camera, battery, display, performance, design, and price contribute to the final score. This explicit approach supports personalization and makes feature trade-offs understandable (Dietz et al., 2022; Chabane et al., 2023).

2.5.3 Cold-Start Handling through Review-Derived Product Profiles

Content and review signals can support recommendation when interaction histories are limited (Volkovs et al., 2017; Lin, 2022; Danish et al., 2026). The proposed system addresses new-user cold-start by collecting feature priorities at the start of the session. New products can be included only after sufficient review evidence is available.

2.5.4 Related Applications beyond Smartphone Recommendation

Studies in healthcare, grocery shopping, tourism, and general e-commerce show that opinion mining, product features, and hybrid models can support decision-oriented recommendation across domains (Keikhosrokiani et al., 2024; Chabane et al., 2023; Dietz et al., 2022; Gajula, 2025). Their methods are transferable, although smartphone-specific vocabulary and evaluation remain necessary.

2.6 Explainable Recommendation and User Trust

2.6.1 Need for Explainability in Smartphone Selection

Smartphones are high-involvement purchases, so users need to understand why a device appears in the recommended list. Explanations can improve perceived usefulness and trust when they reveal the features and trade-offs behind a ranking (Dietz et al., 2022; Chen et al., 2026; Gajula, 2025).

2.6.2 Explanation Generation from Feature Scores

Feature-based explanations can identify the highest weighted contributions and cite the supporting scores or mention counts. Aspect-opinion-sentiment extraction and interpretable ABSA demonstrate how structured evidence can be converted into concise user-facing reasons (Maldonado Castillo et al., 2024; Mansilla et al., 2025; Danish et al., 2026; Zhu, 2025).

2.6.3 Model Interpretability and User-Centered Explanation

Model-centered interpretability describes how a prediction was produced, whereas user-centered explanation states why the recommended product suits the buyer. The proposed system prioritizes the latter by connecting visible feature scores and user

weights to the final ranking, while model-level methods such as causal reasoning or SHAP remain useful for technical inspection (Chen et al., 2026; Zhu, 2025).

2.7 Smartphone and Mobile Review Analysis

Mobile-phone studies provide the closest empirical support for this research. Web-scraped phone reviews contain usable aspect signals, and BERT-based models have outperformed SVM, Naïve Bayes, CNN, RNN, and other classical methods in relevant studies (Soni et al., 2024; Singhal et al., 2025). Multimodal work also shows that emojis and informal expressions can influence aspect sentiment (Padigapati & Krishna, 2025). Related mobile-app studies highlight preprocessing noise, class imbalance, multilinguality, and the need for contextual models (Lestari et al., 2024; Pranatawijaya et al., 2024; Ratniasih et al., 2025). These issues inform the proposed data-cleaning, annotation, model-selection, and evaluation procedures.

2.8 Systematic Literature Review Process

2.8.1 Review Questions

The systematic literature review was conducted to identify how existing studies support the design of an explainable feature-based smartphone recommendation system using ABSA. The review was guided by four questions:

- How can sentiment at the feature level be reliably extracted from smartphone and product reviews?
- How can feature-level sentiment scores be converted into useful product performance metrics?
- How can explainable recommendations be generated using feature scores and user preferences?
- Compared with conventional rating-based systems, how can feature-based recommendation improve interpretability?

2.8.2 Search Strategy and Information Sources

The PRISMA-style search used IEEE Xplore, Springer Nature Link, ResearchGate, and Google Scholar. Searches conducted from 11 to 22 June 2026 covered ABSA, smartphone and product reviews, sentiment-aware and explainable recommendation,

and transformer or machine-learning methods. Final verification was completed on 25 June 2026.

Table 2.1: Search sources, search dates, and records retrieved

Database / Source	Search date	Search fields used	Main filters applied	Records retrieved
IEEE Xplore	22 June 2026	Title, abstract, keywords	English; journal/conference papers; ABSA, sentiment analysis, recommender systems, explainability	86
Springer Nature Link	13 June 2026	Title, abstract, keywords	English; computer science, information systems, AI, NLP, recommendation systems	1,213
ResearchGate	15 June 2026	Title and full-text search	Topic relevance, full-text availability, ABSA/recommendation relevance	53
Google Scholar	18 June 2026	Title and keyword search	2017-2026; relevance to ABSA, product reviews, sentiment-aware recommendation, explainable recommendation	25
Total				1,377

2.8.3 Search Strings

A composite Boolean string combined ABSA and sentiment terms with smartphone or product reviews, recommendation-system terms, and transformer or machine-learning methods. Shorter combinations were used where database interfaces imposed search limitations.

2.8.4 Inclusion and Exclusion Criteria

The criteria retained English academic studies that supported at least one component of the proposed framework. The main publication period was 2021–2026, with selected foundational 2017 studies retained for ABSA, multi-criteria recommendation, and cold-start evidence.

Table 2.2: Inclusion and exclusion criteria

Type	Code	Criteria
Inclusion	IC1	Studies published mainly between 2021 and 2026 were included; foundational earlier works were retained where necessary.
Inclusion	IC2	Studies written in English were included.

Type	Code	Criteria
Inclusion	IC3	Studies related to ABSA, sentiment analysis, opinion mining, product reviews, smartphone reviews, mobile app reviews, or customer reviews were included.
Inclusion	IC4	Studies related to recommender systems, review-aware recommendation, sentiment-aware recommendation, feature-based recommendation, multi-criteria recommendation, cold-start recommendation, or explainable recommendation were included.
Inclusion	IC5	Studies using machine learning, deep learning, BERT, DeBERTa, XLM-RoBERTa, CNN, LSTM, graph-based models, hybrid models, or transformer-based sentiment models were included.
Inclusion	IC6	Studies with accessible full text or enough bibliographic information for DOI/source verification were included.
Inclusion	IC7	Studies that directly or indirectly supported the proposed ABSA-driven smartphone recommendation framework were included.
Exclusion	EC1	Studies unrelated to ABSA, sentiment analysis, user reviews, product reviews, smartphone/mobile reviews, or recommender systems were excluded.
Exclusion	EC2	Duplicate records and repeated studies with the same DOI, title, or author-topic combination were excluded.
Exclusion	EC3	Studies without accessible full text were excluded if their methods and findings could not be verified.
Exclusion	EC4	Non-academic sources such as blogs, marketing websites, general web articles, and unsupported online content were excluded.
Exclusion	EC5	Studies focusing only on general rating prediction without review text, feature-level sentiment, or explainability were excluded.
Exclusion	EC6	Studies outside the scope after full-text eligibility assessment were excluded.

2.8.5 Article Selection Process

The search identified 1,377 records. After duplicate removal, automated and other exclusions, screening, retrieval, and eligibility assessment, 32 studies remained for synthesis.

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only

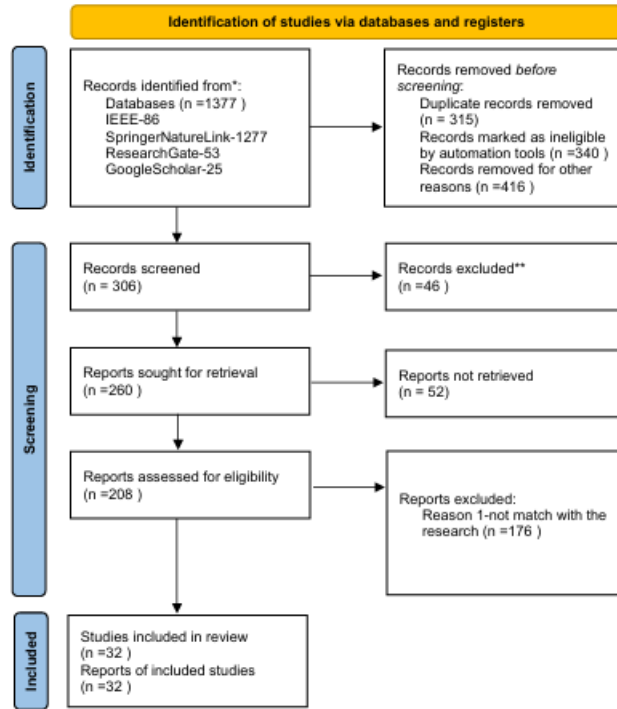


Figure 2.1: PRISMA-style article selection process

2.8.6 Data Extraction and Thematic Coding

Each paper was coded by focus, domain or dataset, method, main findings, limitations, and relevance to review collection, ABSA, scoring, recommendation, explainability, cold-start, multilinguality, or credibility.

Table 2.3: Thematic classification of the selected 32-paper corpus

Theme	Focus	Number of studies
Theme 1	ABSA methods and transformer-based sentiment classification	15
Theme 2	Review-aware and sentiment-aware recommendation	8
Theme 3	Feature-based, multi-criteria, conversational, and cold-start recommendation	5
Theme 4	Smartphone/mobile review evidence, multilinguality, and domain transfer	4
Total		32

2.8.7 Classification of the Selected Papers

Table 2.4 classifies the 32 studies and shows the specific contribution of each paper to the proposed framework.

Table 2.4: Classification of the selected 32 studies

N o.	Study	Main focus	Method / dataset	Relevance to proposed system
1	Musto et al. (2017)	Multi-criteria recommender systems using ABSA	Aspect sentiment from user reviews integrated into multi-criteria collaborative filtering	Justifies converting aspect sentiment into recommendation criteria.
2	Mubarak et al. (2017)	Classical ABSA	POS tagging, Chi-Square feature selection, and Naïve Bayes on product reviews	Provides a baseline for early supervised ABSA methods.
3	Volkovs et al. (2017)	Cold-start recommendation	Content-based neighbor models for recommender system challenge data	Supports content-based signals when interaction history is limited.
4	Liu et al. (2021)	Multilingual review-aware recommendation	MABSA, multilingual embeddings, and dual interactive attention	Shows that aspect sentiment can improve recommendation and interpretability.
5	Dang et al. (2021)	Sentiment-aware recommendation	BERT feature extraction, deep sentiment models, and collaborative filtering	Supports integration of sentiment analysis with recommendation algorithms.
6	Lin (2022)	Cold-start recommendation	User-interaction parallel networks using user, item, social, and review features	Supports the use of review information for cold-start contexts.
7	Dietz et al. (2022)	Feature trade-off recommendation	Trade-off visualization in content-based recommendation	Supports showing feature trade-offs to improve user understanding.
8	Chabane et al. (2023)	Feature-based product recommendation	Clustering and machine learning for grocery recommendation	Shows wider applicability of feature-based product recommendation.
9	Aziz et al. (2024)	Unified transformer ABSA	BERT, graph convolution, biaffine attention, and multiple ABSA subtasks	Supports transformer and graph-enhanced ABSA for aspect extraction and sentiment classification.
10	Rasappan et al. (2024)	E-commerce sentiment classification	Hybrid deep learning with term weighting and feature selection	Supports preprocessing and hybrid deep learning for product reviews.

N o.	Study	Main focus	Method / dataset	Relevance to proposed system
11	Di Nuovo et al. (2024)	Multilingual sentiment analysis	XLM-R-based sentiment model for news data	Supports multilingual transformer adaptation as a future direction.
12	Keikhosrokiani et al. (2024)	Opinion-mining recommendation	Opinion mining and machine learning for drug recommendation	Shows that review sentiment can support decision-oriented recommendation.
13	Lestari et al. (2024)	Mobile app review ABSA	CNN and Word2Vec on Google Play reviews	Highlights mobile-review preprocessing and class imbalance issues.
14	Maldonado Castillo et al. (2024)	Aspect-opinion-sentiment extraction	Triplet extraction and review synthesis for restaurant reviews	Supports generating structured explanations from aspect-opinion-sentiment evidence.
15	Pranatawijaya et al. (2024)	Mobile app review sentiment	Topic modeling, aspect generation, SMOTE, and BERT classification	Supports aspect discovery and class balancing in noisy review data.
16	Sami et al. (2024)	Hybrid recommendation	Deep learning hybrid model using CF, RNN, NCF, and content analysis	Supports combining multiple recommendation signals.
17	Soni et al. (2024)	Smartphone/mobile phone ABSA	Web scraping and ABSA of Flipkart mobile phone reviews; BERT comparison	Directly supports web-scraped mobile phone review ABSA.
18	Gajula (2025)	Sentiment-aware recommendation review	NLP perspective on sentiment-aware e-commerce recommendation	Supports the theoretical value of sentiment in recommendation and explanation.
19	Haghighi and Moradian Zadeh (2025)	Aspect-aware review-based recommendation	DeBERTa aspect-sentiment profiles and interaction modeling	Supports using aspect-sentiment profiles in recommendation.
20	Jayakody et al. (2024)	Instruction-based transformer ABSA	InstructABSA with DeBERTa for aspect extraction and sentiment classification	Supports pipelined aspect extraction and transformer sentiment classification.
21	Kumari et al. (2025)	Product review ABSA	Aspect-based sentiment analysis for product reviews	Supports feature-level product evaluation from customer reviews.

N o.	Study	Main focus	Method / dataset	Relevance to proposed system
22	Mansilla et al. (2025)	Interpretable ABSA	Syntactic patterns, dependency parsing, and VADER sentiment scoring	Supports interpretable aspect-sentiment mapping.
23	Padigapati and Krishna (2025)	Smartphone multimodal ABSA	Text and emoji-based ABSA using Word2Vec, Emoji2Vec, BERT, and BiLSTM	Supports handling informal smartphone review signals.
24	Ratniasih et al. (2025)	Mobile app sentiment review	Review of multilingual and contextual sentiment techniques	Supports transformer shift and multilingual limitations in mobile review analysis.
25	Satria et al. (2025)	Multilingual customer product ABSA	ABAE, FastText/MUSE, and XLM-RoBERTa on multilingual product reviews	Supports multilingual ABSA as a future extension.
26	Setiabudi and Faizah (2025)	Hybrid neural ABSA	BERT embeddings and BiLSTM on FABSA dataset	Supports contextual-sequential ABSA modeling.
27	Singhal et al. (2025)	Mobile phone technical review sentiment	BERT-based sentiment classification compared with classical models	Directly supports mobile phone technical sentiment modeling.
28	Zhu (2025)	Transformer ABSA explainability	DeBERTa-v3, LoRA fine-tuning, and SHAP interpretability	Supports interpretable transformer-based sentiment modeling.
29	Chen et al. (2026)	Explainable recommendation	Causal neuro-symbolic reasoning for multi-behavior recommendation	Supports explainable recommendation logic.
30	Chong et al. (2026)	Multitask ABSA	BERT embeddings, BiLSTM, self-attention, and subjectivity detection	Supports separating subjective review opinions from objective statements.
31	Danish et al. (2026)	Sentiment-aware explainable hybrid recommendation	SBERT, ABSA, collaborative filtering, and SHAP explanations	Supports end-to-end sentiment-aware and explainable recommendation.
32	Jin et al. (2026)	Spam-aware recommendation	Spam review detection and credibility-aware recommendation	Supports review credibility filtering as a quality-control concern.

2.8.8 Data Synthesis

The synthesis links four evidence streams. ABSA research supports aspect and sentiment extraction; recommendation studies support the use of aspect-level evidence in ranking; explainability research supports visible feature-based reasons; and cold-start research supports content and explicit preference signals when user history is limited. Transformer models provide the strongest contextual fit, but the literature also confirms that classification accuracy must be connected to interpretable product scoring and user-facing decision support.

2.9 Critical Comparison of Reviewed Methods

The reviewed methods differ in contextual accuracy, interpretability, computational cost, data requirements, and suitability for recommendation. Classical approaches are efficient and transparent but rely on feature engineering; neural and transformer models capture richer context but require tuning and validation. Review-aware recommendation uses evidence beyond ratings, yet many systems retain latent profiles or model-centered explanations. Table 2.5 summarizes the fit of each method group for this study.

Table 2.5: Critical comparison of reviewed method groups

Method group	Strengths	Limitations	Fit for current research
Lexicon and rule-based ABSA	Interpretable, simple, and able to link aspects to opinion terms	Weak with sarcasm, implicit sentiment, domain-specific language, and noisy reviews	Useful for aspect dictionaries and explanation support, but not sufficient alone.
Classical machine learning	Efficient and workable with smaller labeled datasets	Requires feature engineering and has limited contextual understanding	Useful as baseline but weaker for mixed smartphone review sentences.
Neural and hybrid deep learning	Learns semantic and sequential patterns; can handle richer representations	Requires training data, tuning, and imbalance handling	Useful where transformer embeddings are combined with sequence models.
Transformer-based ABSA	Strong contextual understanding and effective for technical mixed-opinion reviews	Higher computational cost and requires domain adaptation and validation	Most suitable for the core ABSA component of the proposed system.

Method group	Strengths	Limitations	Fit for current research
Review-aware and sentiment-aware recommendation	Uses textual evidence beyond ratings and improves personalization and interpretability	Often focuses on rating prediction or latent profiles rather than transparent scoring	Suitable foundation, but needs explicit user weighting and explanation.
Explainable recommendation	Improves user trust and decision confidence	Some explanations are model-centered or generic	Directly relevant because the proposed system uses feature-score explanations.
Spam-aware and credibility-aware recommendation	Improves reliability of review-based recommendations	Requires labeled spam or credibility data and extra processing	Useful as a data-quality extension for Amazon review analysis.

2.9.1 Comparative Performance of ABSA Models

The evidence shows a progression from Naïve Bayes and engineered features to neural, hybrid, and transformer architectures. Recent BERT- and DeBERTa-based studies provide the strongest contextual performance for multi-aspect and technical review language (Mubarok et al., 2017; Aziz et al., 2024; Jayakody et al., 2024; Soni et al., 2024; Singhal et al., 2025; Chong et al., 2026).

2.9.2 Limitations of Existing Approaches

Three limitations remain: ABSA studies often stop at classification; recommender studies frequently prioritize rating prediction or latent interactions; and explanation methods may not directly connect review evidence, user priorities, and product rank. These limitations define the integration requirement of the proposed system.

2.10 Research Gap and Positioning of the Study

The selected literature establishes ABSA, transformer sentiment classification, review-aware recommendation, and explainability, but their full smartphone-specific integration remains limited. Mobile-phone studies extract useful feature sentiment, yet they do not generally build a transparent user-weighted smartphone ranking engine (Soni et al., 2024; Padigapati & Krishna, 2025; Singhal et al., 2025).

A second gap concerns preference representation. Aspect-aware recommenders show that feature sentiment can improve prediction, but many rely on collaborative filtering,

learned profiles, or neural interactions (Musto et al., 2017; Liu et al., 2021; Haghighi & Moradian Zadeh, 2025). The proposed system instead uses explicit feature weights and an auditable scoring formula.

A third gap is user-interface explanation. Existing work provides syntactic, attribution, trade-off, or neuro-symbolic explanations, but fewer systems state in simple language why a particular smartphone matches the user’s priorities using actual feature scores (Dietz et al., 2022; Mansilla et al., 2025; Zhu, 2025; Chen et al., 2026).

The study is therefore positioned as an applied design-and-development project integrating Amazon review collection, preprocessing, ABSA, feature scoring, user weighting, ranking, and evidence-based explanation in one smartphone prototype.

Table 2.6: Research gaps and positioning of the proposed study

Identified gap	Evidence from selected literature	How the proposed study addresses it
Gap 1: Smartphone specificity	Most ABSA and recommendation studies use restaurants, laptops, apps, books, movies, groceries, healthcare, tourism, or general e-commerce datasets. Smartphone-specific ABSA-recommendation integration remains limited.	Uses Amazon smartphone reviews and predefined smartphone features: camera, battery, display, performance, design, and price.
Gap 2: User-weighted feature scoring	Many systems use latent profiles, ratings, collaborative filtering, or neural interaction models rather than explicit user feature priorities.	Uses a transparent weighted scoring formula where user-selected feature weights are multiplied by review-derived feature scores.
Gap 3: Explanation generation	Existing explanations are often model-centered, generic, or not linked to actual feature sentiment percentages.	Generates plain-language explanations grounded in feature sentiment scores and user-selected priorities.
Gap 4: End-to-end integration	ABSA, recommendation, cold-start handling, data scraping, scoring, and explainability are often studied separately.	Integrates Amazon scraping, preprocessing, ABSA, feature scoring, user-weighted ranking, and explanation generation in one prototype.

2.12 Chapter Summary

The review confirms that ABSA and transformer models can recover feature-level opinion from noisy product text, while review-aware, multi-criteria, cold-start, and explainability studies support explicit feature evidence and visible trade-offs in

recommendation. The remaining gap is a cohesive smartphone-specific system that converts Amazon reviews into transparent feature scores, user-weighted rankings, and plain-language explanations. The conceptual framework addresses this gap and guides the methodology in Chapter 3.

Chapter 3 : Methodology

3.1 Chapter Introduction

This chapter presents the methodology used to develop and evaluate the explainable feature-based smartphone recommendation system. The methodology is reported according to the system that was implemented and the data that were actually processed. The study uses a frozen offline dataset of publicly visible Amazon smartphone reviews obtained through Bright Data and stored locally for reproducible analysis. Review evidence is processed through a sequential pipeline that screens eligible smartphone listings, cleans and filters reviews, segments retained text, detects six smartphone aspects, assigns aspect-level sentiment, aggregates the resulting evidence into product-feature scores, applies user-defined feature priorities, ranks eligible smartphones, and produces evidence-grounded explanations.

The six aspects are battery, camera, display, performance, design, and price or value. They are derived from language in customer reviews rather than copied from manufacturer specifications. Each detected aspect mention is classified as positive, neutral, or negative. These predictions are retained as auditable intermediate records before being aggregated into feature scores. A Bayesian shrinkage procedure reduces the influence of very small evidence counts, while explicit user weights determine how strongly each feature contributes to a final recommendation score. The proposed ranking is compared with a conventional Amazon average-star-rating baseline. The overall methodological direction is informed by prior work on smartphone review analysis, sentiment-aware recommendation, and explainable aspect-based models (Soni et al., 2024; Musto et al., 2017; Haghighi & Moradian Zadeh, 2025).

The implemented primary Aspect-Based Sentiment Analysis (ABSA) method is a versioned domain lexicon supported by contextual rules for negation, intensification, diminution, and contrast. An OpenAI-compatible large language model interface is available as an optional secondary annotation path, but it is not required for the reproducible offline pipeline. The methodology therefore does not claim that a BERT model was trained, does not use a 70:15:15 transformer training split, and does not depend on a large manually gold-labelled dataset. Evaluation focuses on the verified data funnel, aspect and sentiment coverage, star-text consistency as a sanity check,

ranking behavior under predefined user-priority profiles, comparison with the Amazon-rating baseline, and numerical faithfulness of explanations.

Accordingly, the chapter documents the complete transformation of data from the raw review file to the final ranking and evaluation outputs. This process-oriented description establishes how each output is produced, stored, checked, and passed to the next analytical stage.

3.2 Research Philosophy

The study adopts a pragmatic research philosophy. Pragmatism is appropriate because the methods are selected according to their ability to solve a practical smartphone decision-support problem. The research does not assume that one analytical technique provides a complete representation of smartphone quality. Aggregate product ratings provide a familiar baseline, while review text contains more detailed evidence about individual features. The methodology combines these forms of evidence in a way that can be implemented, inspected, evaluated, and reproduced.

Under this position, Amazon review text is treated as observable user-generated evidence. The text is transformed through natural language processing into aspect-sentiment records, feature scores, weighted contributions, final recommendation scores, ranks, and explanations. The value of the methodology is assessed through the usefulness and consistency of these outputs rather than through the testing of a causal social-science model.

Pragmatism also supports iterative artefact development followed by analysis on a frozen data snapshot. During development, preprocessing rules, aspect vocabulary, sentiment rules, aggregation parameters, and explanation templates were refined to address observed data-quality and language problems. Once the principal workflow was established, the corpus was stored locally and reused for analysis so that the reported counts and ranking behavior were not affected by changes to live Amazon pages.

3.3 Research Approach and Type of Study

The research is predominantly quantitative and computational. Unstructured review text is converted into categorical and numerical outputs: aspect categories, sentiment labels, mention counts, raw feature scores, Bayesian-smoothed scores, normalized user weights, weighted feature contributions, final recommendation scores, and ranks.

Descriptive statistics are then used to explain the composition of the corpus and the distribution of the analytical outputs.

The study is applied research because it develops a working decision-support artefact for smartphone selection. It is also a design-and-development study because the contribution lies in integrating data preparation, ABSA, evidence aggregation, preference weighting, ranking, explanation, and evaluation into one traceable workflow. The study is not a survey, interview study, or causal hypothesis-testing investigation. Human participants are not required for the core methodology, and no questionnaire data are used to produce the reported corpus or recommendation results. The quantitative approach does not imply that all outputs are ground truth. Aspect and sentiment records are automated predictions, and feature scores are computational summaries of the available review evidence. The methodology therefore reports evidence volume, processing decisions, provenance, missing evidence, and limitations together with the numerical scores.

3.4 Research Strategy

An applied design-and-development strategy is combined with controlled technical evaluation. The artefact was developed in modules so that the output of each stage could be inspected before it became the input to the next stage. The principal stages are data acquisition, product eligibility screening, database ingest, review preprocessing, sentence and clause-aware segmentation, aspect identification, aspect-level sentiment classification, feature-score aggregation, user-weighted ranking, explanation generation, and evaluation.

Development was iterative because the raw dataset contained non-phone listings, multilingual text, duplicated or suspicious content, very short comments, uneven numbers of reviews per listing, and feature language that varied between reviewers. These conditions required refinements to product filters, text-quality rules, aspect terms, sentiment vocabulary, and scoring safeguards. The final methodological description, however, refers to the implemented pipeline and the frozen corpus rather than to every preliminary experiment.

The strategy maintains separation between source evidence and derived outputs. Raw review rows are retained separately from normalized text, segmented sentences, aspect-sentiment predictions, aggregate scores, and recommendation results. This separation

allows a feature score or explanation to be traced backwards to the processed evidence from which it was calculated.

3.5 Research Design and Time Horizon

The research follows a sequential and modular design. Each stage has a defined input, transformation, output, and storage location. Figure 3.1 summarizes the principal flow. The process starts with the Bright Data Amazon review export and finishes with ranked smartphones, traceable explanations, baseline comparisons, and evaluation summaries.

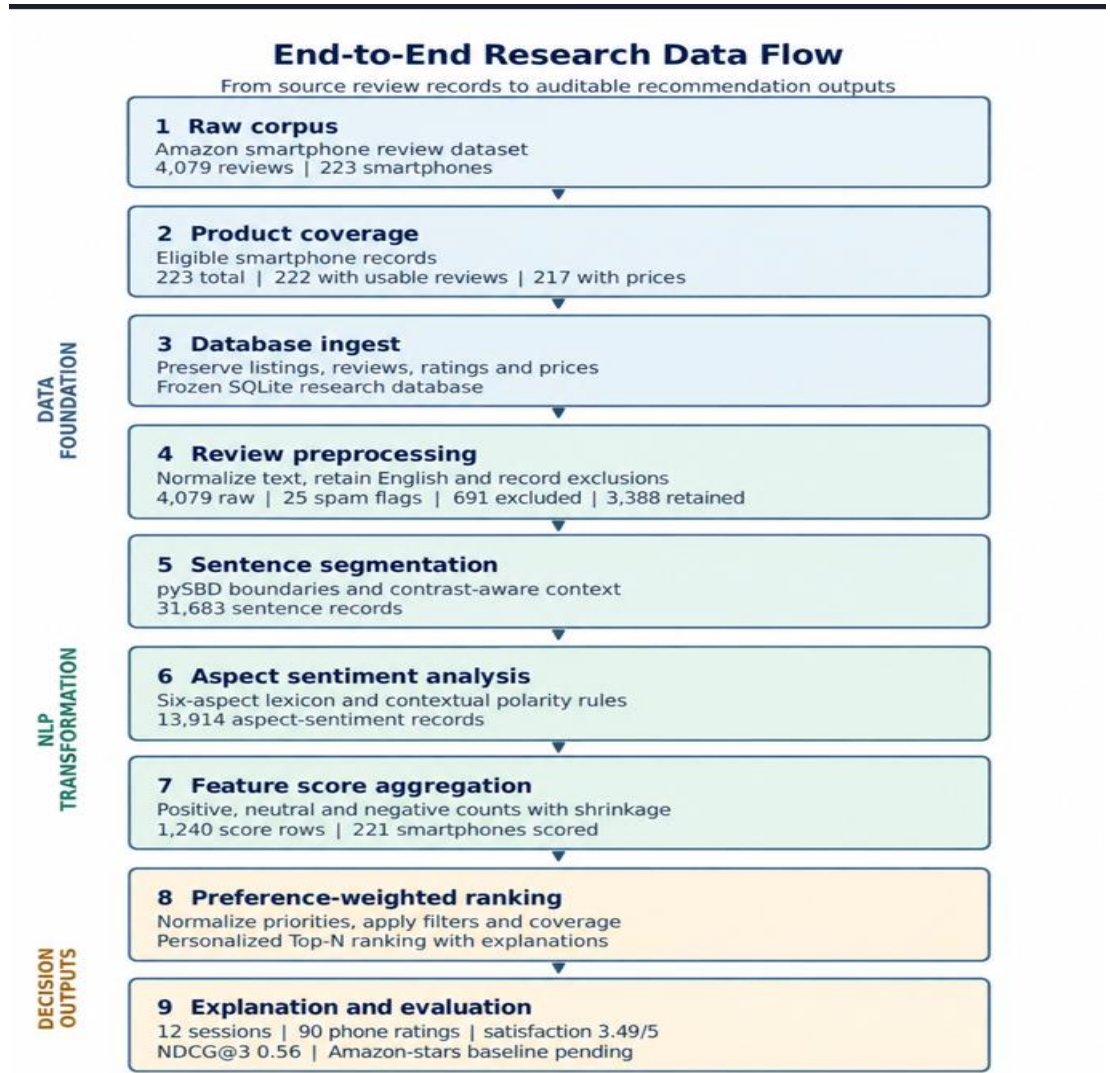


Figure 3.1: End-to-end research data flow from the raw Amazon review corpus to recommendation and evaluation outputs. Source: Developed by the researcher.

The time horizon is cross-sectional. The reviews represent a documented snapshot of Amazon content rather than a continuously updated longitudinal dataset. The raw file and database are frozen for the reported analysis. Consequently, the results describe the evidence available in that snapshot and should not be interpreted as evidence of changes in smartphone performance over time.

Table 3.1 summarizes the overall methodological profile.

Table 3.1: Overall methodological profile

Methodological component	Adopted choice	Reason for the choice
Research philosophy	Pragmatism	Prioritizes methods that produce a useful, auditable smartphone decision-support artefact.
Research approach	Predominantly quantitative and computational	Converts review text into measurable labels, evidence counts, scores, rankings, and checks.
Research type	Applied design-and-development research	Develops and evaluates an implemented solution to a practical selection problem.
Research strategy	Modular artefact development with controlled technical evaluation	Allows each transformation to be verified independently and as part of the complete pipeline.
Time horizon	Cross-sectional	Uses one frozen snapshot of Amazon review evidence.
Main data source	Secondary user-generated Amazon review data obtained through Bright Data	Provides feature-specific product experiences without recruiting participants.
Primary ABSA method	Versioned domain lexicon with contextual sentiment rules	Supports offline execution, transparency, and reproducibility.
Comparison method	Amazon average-star-rating ranking	Provides a familiar non-personalized baseline.
Human participants	Not required for the core methodology	Corpus analysis, ranking comparison, and explanation checks can be completed without a recruited study.

The relationship between the research objectives and the implemented procedures is shown in Table 3.2.

Table 3.2: Alignment between research objectives and methodological procedures

Research objective	Implemented methodological procedure	Principal output
RO1: Prepare an Amazon smartphone review dataset	Validate the Bright Data export, remove non-phone listings, load eligible records, filter unusable reviews, and preserve identifiers and processing status.	Frozen, traceable corpus and verified data funnel.
RO2: Identify smartphone features and classify aspect sentiment	Apply a closed six-aspect vocabulary and contextual lexicon rules, with an optional LLM annotation path when configured.	Aspect-sentiment records linked to source sentences and listings.
RO3: Generate feature-performance scores	Aggregate positive, neutral, and negative predictions and apply Bayesian shrinkage using evidence counts.	Comparable product-aspect scores with mention counts and coverage information.
RO4: Rank smartphones according to user priorities	Normalize feature weights, apply optional price filters, impute missing candidate values with explicit flags, and calculate weighted totals.	Personalized Top-N ranking and Amazon-star baseline ranking.
RO5: Produce explainable recommendations	Derive reasons from the same feature contributions used in the final score and disclose low coverage or important weaknesses.	Traceable explanation for each ranked product.

3.6 Population Units of Analysis and Actual Sample

3.6.1 Data Population

The target data population comprises publicly visible English-language customer reviews associated with smartphone listings on Amazon. The accessible population is bounded by the records present in the Bright Data Amazon cell-phone or smartphone review export used for the study. Reviews are secondary data because they were written for product feedback and purchasing decisions rather than specifically for this research. The population does not include manufacturer descriptions, professional review articles, social-media comments, private account information, or reviews not present in the frozen source file. Accessories, manuals, guidebooks, tablets, smartwatches,

laptops, and unrelated electronics are outside the product boundary. Reviewer names and profile links are not needed for the analysis and are excluded from the analytical outputs.

3.6.2 Units of Analysis

Several units are distinguished because the pipeline operates at different levels. The raw review row is the initial data-record unit. After preprocessing, the retained review is the document unit. Sentence segmentation creates the principal natural language processing unit. Where contrastive language separates opinions, the lexicon engine evaluates local clauses or contexts so that sentiment is not transferred incorrectly between aspects. The aspect-sentiment mention is the immediate ABSA output. The smartphone listing, identified through an Amazon Standard Identification Number (ASIN), is the aggregation and recommendation unit.

For example, the review statement “The camera is excellent, but the battery drains quickly” contains two distinct opinions. The sentence should produce a positive camera record and a negative battery record. Treating the whole sentence as one sentiment label would remove the distinction needed for feature-based recommendation.

An ASIN identifies a listing rather than guaranteeing a unique canonical smartphone model. Different configurations or duplicated marketplace listings may therefore remain separate candidates when they have separate identifiers and review pools. For this reason, the methodology reports ASIN-associated smartphone listings and does not claim that every identifier represents a completely unique hardware model.

3.6.3 Actual Computational Sample

The computational sample consists of the records that passed the product and review eligibility procedures. The stored dataset contained 4,079 review rows associated with 223 eligible smartphone listings. Preprocessing retained 3,388 usable English reviews and excluded 691 records. Spam-oriented heuristics marked 25 reviews during preprocessing; these flags overlap with the applied exclusion decisions and are not added as a separate category. The retained reviews produced 31,683 segmented sentences and 13,914 aspect-sentiment mentions. Aggregation generated 1,240 product-aspect score rows, and 221 listings received at least one feature score.

Table 3.3: Verified dataset funnel

Processing stage	Unit retained	Verified size
Raw reviews stored	Review rows and eligible smartphones	4,079 reviews and 223 smartphones
Spam flag status	Reviews marked by spam heuristics	25 reviews (overlapping exclusion decisions)
Loaded into the research database	Smartphone listings and reviews	223 listings and 4,079 reviews
Step 1 preprocessing	Usable English reviews	3,388 reviews
Step 1 exclusions	Removed review records	691 reviews
Step 2 segmentation	Sentences from retained reviews	31,683 sentences
Steps 3 and 4 ABSA	Aspect-sentiment mentions	13,914 records
Step 6 aggregation	Product-aspect score rows	1,240 rows
Final score coverage	Listings with at least one aspect score	221 listings

Among the 223 listings with at least one retained review, the number of retained reviews ranged from 1 to 24, with a mean of approximately 15.2 reviews per listing. The dataset therefore provides broad product coverage but limited evidence depth for some listings. The implemented methodology retains these listings and addresses evidence uncertainty through mention counts, Bayesian shrinkage, coverage indicators, and low-evidence flags rather than claiming that every feature score has equal reliability.

The 691 preprocessing exclusions are detailed in Table 3.4. Spam-oriented heuristics flagged 25 reviews during preprocessing; these flags overlap with the applied exclusion decisions and are not added as a separate category to the total.

Table 3.4: Verified preprocessing exclusions

Exclusion reason	Count	Percentage of all exclusions
Non-English language	663	95.95%
All-capital text	12	1.74%
No alphabetic content	10	1.45%
Extremely short text	3	0.43%

Exclusion reason	Count	Percentage of all exclusions
Generic-only content	3	0.43%
Total	691	100.00%

3.7 Sampling Strategy and Eligibility Criteria

The study applies criterion-based census sampling within the frozen source file. It does not draw a random sample from all Amazon reviews. Instead, every source record available in the snapshot is considered and is retained when it satisfies the product and review eligibility rules. This approach is appropriate because the objective is to construct and examine the implemented pipeline using all eligible evidence available in the acquired corpus.

3.7.1 Product Listing Eligibility

Product eligibility is assessed before text analysis. A record must be linked to an ASIN and represent a smartphone listing. Titles that indicate manuals, handbooks, guidebooks, accessories, or unrelated products are removed. The final stored dataset contained 223 eligible smartphone listings.

3.7.2 Review Eligibility

Eligible reviews must contain meaningful textual content, remain linked to an eligible listing, and fall within the English-language scope. Empty text, text without alphabetic characters, extremely short unusable comments, generic-only content, and records rejected by quality controls are excluded. Exact duplication and suspicious patterns are checked to reduce artificial inflation of aspect counts because misleading or repeated reviews can distort recommendation evidence (Jin et al., 2026).

Table 3.5: Inclusion and exclusion criteria

Code	Inclusion criterion	Code	Exclusion criterion
IC1	Record is linked to an eligible Amazon smartphone ASIN.	EC1	Record concerns a manual, guidebook, accessory, tablet, wearable, or unrelated product.
IC2	Review contains meaningful textual content.	EC2	Review is empty, contains no alphabetic content, or is too short to provide usable evidence.

Code	Inclusion criterion	Code	Exclusion criterion
IC3	Review is classified as English for the main analysis.	EC3	Review is classified as non-English.
IC4	Review can be traced to a source product and database record.	EC4	Product identity or source linkage cannot be established.
IC5	Review remains after text-quality and duplicate controls.	EC5	Review is an exact duplicate, generic-only record, or unusable suspicious pattern.
IC6	Retained text can proceed to segmentation and ABSA.	EC6	Processing fails or no valid text remains after normalization.

3.7.3 Treatment of Uneven Review Evidence

The number of reviews is not balanced across listings, and aspect mentions are not balanced within a listing. This imbalance reflects the available marketplace evidence rather than a designed experimental sample. Artificially forcing an equal number of reviews or aspect mentions would discard source evidence and could create a misleadingly uniform dataset.

The methodology instead preserves the observed distribution and controls its effect at the scoring stage. Each stored feature score is accompanied by its positive, neutral, and negative counts. Bayesian shrinkage moves scores supported by few mentions toward the aspect-wide mean. A minimum-evidence setting identifies low-confidence scores, and coverage indicators show how much of the requested recommendation weight is supported by observed mentions. These controls reduce, but do not eliminate, uncertainty caused by sparse or uneven evidence.

3.8 Data Acquisition Ingest and Storage

3.8.1 Source and Acquisition Mode

The principal source is a Bright Data Amazon cell-phone or smartphone review export delivered as a structured dataset and stored locally as `data/full_reviews.csv`. Using an offline file separates the analytical process from live Amazon access, prevents page changes during an experiment from altering the results, and makes the reported analysis reproducible.

The implementation also contains optional collection utilities based on Playwright, BeautifulSoup, lxml, and retry controls. These utilities can support supplementary acquisition experiments, but live scraping is not required for the principal research workflow and is not performed during each recommendation request.

3.8.2 Data Ingest Procedure

The ingest procedure follows six stages. First, the original export is preserved as the raw source snapshot. Second, required fields and product identifiers are validated. Third, manual and guidebook title patterns are removed. Fourth, smartphone, review, price, and rating metadata are loaded into the SQLite research database through SQLAlchemy models. Fifth, preprocessing and downstream analytical jobs operate on the imported records without overwriting the original review text. Finally, processing counts and status information are exported for verification.

Table 3.6: Principal source fields and their methodological use

Source field	Description	Use in the research
ASIN or product identifier	Amazon listing identifier	Groups reviews and preserves listing-level traceability.
Product name and brand	Displayed listing description	Supports eligibility checks and identification of recommendation candidates.
Product URL	Source listing address when available	Supports audit and source tracking.
Listing price	Displayed price at collection time when available	Supports optional budget filtering; it is not treated as sentiment.
Amazon average rating	Aggregate star rating shown for the listing	Provides the conventional baseline ranking.
Review title and body	Written customer feedback	Provides the primary text for preprocessing and ABSA.
Individual review rating	Star value associated with a review	Supports descriptive reporting and star-text consistency checks.
Review date and available metadata	Timing and source context	Supports dataset description and traceability where present.

3.8.3 Storage and Traceability

SQLite is used as the local research database, and SQLAlchemy 2.0 provides the data-access layer. Source and derived records are stored separately. Principal entities include smartphone listings, reviews, segmented sentences, aspect-sentiment predictions, product-aspect scores, price observations, recommendation sessions, and evaluation feedback where a demonstration session is recorded.

Traceability is preserved through stable internal identifiers and relationships. A product-aspect score can be traced to its aggregated aspect records, which can be traced to sentences and source reviews. Engine provenance is stored with ABSA outputs so that a lexicon prediction can be distinguished from an optional LLM annotation or a lexicon fallback. Processing jobs record their status and configuration so failures are visible rather than silently omitted.

3.8.4 Data Security

The analysis does not require reviewer names, profile links, account credentials, or other direct personal identifiers. Raw and processed files are stored locally within the research project. Environment variables are used for optional external-service credentials rather than embedding keys in source code. The raw source file is preserved separately from cleaned and derived tables so that analytical transformations do not alter the original evidence.

3.9 Review Preprocessing and Segmentation

3.9.1 Step 1- Review Preprocessing

Preprocessing is deliberately conservative because punctuation, negation, intensifiers, and contrast terms can change sentiment meaning. The pipeline removes technical noise but retains the language required for aspect and polarity decisions. The implementation validates product linkage, removes HTML remnants, repairs common encoding artefacts, applies Unicode normalization, collapses repeated whitespace, checks language, detects unusable text, and applies duplicate and spam-oriented controls. These controls respond to the duplication, imbalance, and preprocessing risks identified in mobile-review and e-commerce sentiment research (Rasappan et al., 2024; Lestari et al., 2024).

English-language filtering is performed using langdetect. Language classification is a major transformation in this corpus, accounting for 663 of the 691 excluded reviews. Because automatic language detection can misclassify short or mixed-language records, the exclusion count is retained and reported as a methodological limitation. The all-capital rule also requires caution because capitalization alone does not necessarily make an opinion unusable; this rule should therefore remain auditable in future revisions. The original text is not overwritten. Normalized text and exclusion reasons are stored separately so that the decision applied to each review can be reconstructed.

3.9.2 Step 2- Sentence and Clause-Aware Segmentation

Sentence segmentation is performed with pySBD, with fallback handling for short or irregular fragments. The 3,388 retained reviews produced 31,683 sentences, equivalent to approximately 9.35 sentences per retained review. Segmentation reduces the amount of unrelated context passed to the ABSA engine and improves the link between a detected aspect and the sentiment expressed toward it.

The lexicon engine also evaluates local contexts around aspect terms and treats contrast markers such as *but*, *however*, *although*, and *while* as boundaries between potentially opposing opinions. This does not create a separately stored clause corpus in every case; it is a context-control procedure used during sentiment calculation. The purpose is to prevent a positive term attached to one aspect from being applied to a different aspect elsewhere in the sentence.

3.9.3 Preprocessing Outputs

Table 3.7: Implemented data-preparation procedures

Procedure	Implementation	Stored or analytical output
Product validation	Confirm ASIN and eligible smartphone listing	Traceable candidate record.
Non-phone screening	Remove manual and guidebook title patterns	Smartphone-only import set.
Text normalization	Strip HTML remnants, normalize Unicode, repair encoding, and collapse whitespace	Clean text while preserving the source field.

Procedure	Implementation	Stored or analytical output
Language filtering	Retain English reviews using langdetect	Language decision and exclusion reason.
Quality controls	Check empty, no-alphabetic, extremely short, generic-only, all-capital, duplicate, and suspicious patterns	Retention status and auditable reason.
Sentence segmentation	Apply pySBD boundaries with fallback rules	Sentence records linked to reviews.
Context handling	Use local windows and contrast markers during sentiment scoring	Reduced sentiment leakage between aspects.

3.10 Aspect Based Sentiment Analysis

3.10.1 Closed Aspect Taxonomy

The ABSA process uses a closed taxonomy of six aspects selected because they represent common smartphone decision criteria and align with the research objectives. The same taxonomy is used during aspect detection, aggregation, weighting, ranking, explanation, and reporting. Maintaining one versioned aspect set avoids category drift between pipeline stages.

Table 3.8: Smartphone aspect taxonomy

Aspect	Indicative language	Operational meaning
Battery	battery, charge, charging, drain, power, backup, fast charging	Endurance, charging behaviour, and power reliability.
Camera	camera, photo, image, video, selfie, lens, zoom, portrait, night mode	Image and video quality and shooting experience.
Display	display, screen, brightness, resolution, AMOLED, color, touch, refresh rate	Visual quality, readability, smoothness, and touch experience.
Performance	speed, lag, processor, RAM, gaming, heat, responsiveness, multitasking	Responsiveness, stability, thermal behavior, and processing experience.

Aspect	Indicative language	Operational meaning
Design	design, build, body, look, weight, grip, finish, premium, durability	Appearance, ergonomics, materials, and physical construction.
Price or value	price, cost, value, worth, expensive, cheap, budget, deal, money	Affordability and perceived value for money.

The vocabulary includes domain synonyms and phrase variants. Longer phrases are evaluated before shorter tokens where necessary to reduce partial matches. The lexicon is version controlled so that changes can be connected to a specific analysis run.

3.10.2 Aspect Identification

Aspect identification locates explicit feature terms in each segmented sentence. One sentence can produce more than one aspect record when several features are mentioned. The method does not force the complete sentence into one category. Each detected aspect retains the sentence and review relationship required for later inspection.

The closed lexicon improves transparency because the reason for an aspect match can be inspected. Its main limitation is that highly implicit opinions may be missed. For example, “it lasts all day” may refer to battery endurance without containing the word *battery*. Phrase patterns can recover some implicit cases, but the methodology does not assume complete coverage of every possible expression.

3.10.3 Aspect-Level Sentiment Classification

The primary sentiment engine assigns positive, neutral, or negative polarity to each detected aspect. It combines a domain sentiment lexicon with local contextual rules. Positive and negative opinion terms provide the initial polarity. Negation windows reverse or modify the polarity of nearby terms, while intensifiers and diminishers adjust strength. Contrast markers restrict the context that can influence an aspect. If an aspect is detected without a sufficiently clear evaluative cue, the record is classified as neutral.

The three-class scheme is operationalized as follows:

- **Positive:** the local context expresses satisfaction, strength, benefit, or favorable value toward the aspect.
- **Negative:** the local context expresses dissatisfaction, weakness, failure, or unfavorable value toward the aspect.

- **Neutral:** the aspect is described or mentioned without a clear favorable or unfavorable judgement.

Opposing opinions are represented as separate aspect records where possible. A fourth *mixed* class is not used because it would weaken aggregation and conceal which aspect received which polarity.

3.10.4 Optional LLM Annotation Path

The implementation supports an optional OpenAI-compatible HTTP annotation service. When configured, sentences can be submitted in batches and returned as structured JSON containing aspect and sentiment labels. Automatic mode may use the external engine when credentials and connectivity are available and return to the lexicon engine when the external path is unavailable. Provenance is recorded for every output. This optional capability is not described as a locally trained transformer model. It does not involve fine-tuning BERT, creating a 70:15:15 training split, or claiming a test-set Macro F1 score. The lexicon engine remains the principal reproducible method because it can execute offline with fixed rules and no external service dependency.

3.10.5 ABSA Output

The pipeline produced 13,914 aspect-sentiment mentions from the retained sentences. Each record includes the product, review or sentence linkage, aspect, sentiment, confidence or method-related information where available, and engine provenance. These records form the evidence base for feature-score aggregation. They are automated analytical annotations rather than independently verified manual labels; this distinction is maintained when reporting results.

3.11 Feature Score Generation

3.11.1 Sentiment Aggregation

ABSA records are grouped by smartphone listing (p) and aspect (a). Let ($P_{p,a}$), ($U_{p,a}$), and ($N_{p,a}$) denote the numbers of positive, neutral, and negative mentions. The total evidence count is:

$$n_{p,a} = P_{p,a} + U_{p,a} + N_{p,a}$$

The implemented raw feature score is:

$$\text{RawScore}_{p,a} = \frac{P_{p,a} + 0.5U_{p,a}}{P_{p,a} + U_{p,a} + N_{p,a}}$$

The score ranges from zero to one. Positive evidence receives a value of one, neutral evidence contributes 0.5, and negative evidence contributes zero. Neutral mentions therefore represent an intermediate contribution rather than being discarded or treated as positive. Positive, neutral, and negative counts are retained with the score so that the numerical value can be interpreted together with its evidence volume.

3.11.2 Bayesian Shrinkage

A raw proportion can appear extreme when it is supported by only one or two mentions. To reduce this problem, the stored ranking score applies Bayesian shrinkage toward the mean score for the relevant aspect. Let (μ_a) represent the corpus or candidate prior mean for aspect (a), and let (k) be the configured shrinkage strength. The implemented score is:

$$\text{Score}_{p,a} = \frac{n_{p,a} \text{RawScore}_{p,a} + k\mu_a}{n_{p,a} + k}$$

The default value of (k) is 5. As evidence increases, the observed raw score receives greater influence. When evidence is sparse, the score remains closer to the prior mean. Shrinkage improves comparative stability but does not remove product-selection bias, listing duplication, missing aspects, or errors in the underlying ABSA predictions.

3.11.3 Evidence Status and Coverage

The aggregation stage generated 1,240 product-aspect score rows for 221 listings. A complete six-aspect matrix for 221 listings would contain 1,326 rows; therefore, some product-aspect combinations have no detected evidence. Design evidence was available for 200 listings, illustrating that coverage differs by feature.

Each score is accompanied by its mention count and evidence status. The default minimum-mention setting is three. This value identifies low-confidence scores; it should not be interpreted as proof that three mentions are statistically sufficient. Evidence counts, score coverage, and imputation flags must therefore remain visible during interpretation.

3.12 Preference Weighting Recommendation and Explanation

3.12.1 User Preference Weights

The recommender accepts importance values for battery, camera, display, performance, design, and price or value. The values are normalized before scoring. For entered weight (w_a), the normalized weight is:

$$W_a = \frac{w_a}{\sum_{j=1}^6 w_j}$$

If all entered values are zero, equal weights are applied. This safeguard prevents division by zero and provides a neutral default profile. The ranking calculation uses the normalized values, so changing the scale of all entered weights without changing their relative proportions does not change the result.

3.12.2 Candidate Selection and Missing Evidence

Optional minimum and maximum price filters may restrict the candidate set using observed listing prices. Price filtering is separate from the review-derived price or value aspect: the first represents a displayed numeric price, while the second represents reviewers' opinions about affordability or value.

Not every candidate has a score for every requested aspect. The implementation imputes a missing aspect with the mean score of the current candidate set and records an imputation flag. This prevents a missing value from being treated automatically as a negative score. It also prevents a candidate from disappearing solely because reviewers did not discuss one feature. Because mean imputation reduces differentiation and can conceal uncertainty, the output includes coverage measures and flags. A stricter future evaluation may exclude products that lack real evidence for the user's most heavily weighted features.

3.12.3 Weighted Recommendation Score

For each eligible listing, the proposed recommendation score is:

$$\text{RecommendationScore}_p = \sum_{a=1}^6 W_a \text{Score}_{p,a}$$

Each term ($W_a \text{Score}_{p,a}$) is stored or reconstructable as the contribution of aspect (a) to product (p). Candidates are sorted from the highest to the lowest total score, and

the requested Top-N products are returned. Because the calculation is additive and explicit, a ranking can be reconstructed from the feature scores and normalized weights. This design follows the principle that review-derived product profiles and explicit user priorities can support transparent feature-based recommendation (Musto et al., 2017; Haghighi & Moradian Zadeh, 2025).

Weighted evidence coverage is calculated as the proportion of the user’s requested weight supported by observed rather than imputed feature evidence. A product can therefore have a high recommendation score but a lower coverage value, signaling that the result should be interpreted cautiously.

3.12.4 Amazon Star-Rating Baseline

The comparison method ranks the same eligible candidate pool by Amazon average product rating. It does not apply user feature weights and does not use feature-specific review evidence. The baseline answers which products have the highest aggregate star ratings, whereas the proposed method answers which products best match the stated feature priorities according to the processed review evidence.

Using the same candidate pool and price filters prevents differences caused only by comparing different products. Differences in Top-N membership or order can then be attributed to the ranking methods, subject to the limitations of the underlying data.

3.12.5 Explanation Generation

Explanations are generated from the same scores, weights, contributions, evidence counts, and coverage information used in ranking. The explanation module identifies the largest weighted contributions as the principal reasons for a recommendation. It may also report a lower-performing heavily weighted feature, limited evidence, or an imputed aspect where relevant. An explanation for the baseline states that the order is based on Amazon average rating rather than implying that feature evidence determined the position.

Rule-based generation is used to preserve computational faithfulness. The explanation should not introduce specifications, marketing claims, or reasons that are absent from the stored calculation. This design makes the explanation inspectable and allows automated consistency checks between the displayed reason and the numerical ranking inputs, consistent with the emphasis on traceable aspect evidence in explainable recommendation research (Mansilla et al., 2025; Dietz et al., 2022).

3.13 End to End Data Flow and Analytical Outputs

The complete research flow is presented in Table 3.9. It specifies how the raw evidence becomes a recommendation. Every stage creates an output that can be inspected independently and used by the next stage.

Table 3.9: End-to-end data flow and stage outputs

Stage	Input	Principal transformation	Output passed forward
1	Bright Data Amazon CSV with 4,079 stored reviews linked to 223 smartphones	Validate required fields, confirm eligible smartphones, and apply spam and text-quality checks	4,079 stored reviews linked to 223 smartphone listings.
2	Stored source rows	Load listing, review, price, and rating data into SQLite while preserving source text	Frozen research database with 223 listings and 4,079 reviews.
3	Imported reviews	Normalize text, filter language, remove unusable records, and record exclusion reasons	3,388 retained English reviews and 691 exclusions.
4	Retained reviews	Apply sentence boundaries and context-aware handling of contrastive clauses	31,683 linked sentence records.
5	Segmented sentences	Detect the six aspects and classify local sentiment using the lexicon engine or optional LLM path	13,914 aspect-sentiment records with method provenance.
6	Aspect-sentiment records	Count positive, neutral, and negative evidence and calculate raw and Bayesian-smoothed scores	1,240 product-aspect score rows covering 221 listings.
7	Candidate feature profiles and user priorities	Normalize weights, apply price filters, manage missing evidence, calculate contributions, and sum scores	Personalized recommendation scores, coverage indicators, and Top-N ranking.

Stage	Input	Principal transformation	Output passed forward
8	Ranked products and stored contributions	Select strongest supported contributions and relevant weaknesses	Plain-language evidence-grounded explanations.
9	Proposed and baseline results	Compare methods, inspect coverage and consistency, and calculate available evaluation summaries	Corpus, ABSA, ranking, robustness, and explanation-evaluation outputs.

This flow preserves three important distinctions. First, raw review rows are not equivalent to retained reviews. Second, sentences and aspect-sentiment mentions are derived analytical units rather than independent respondents. Third, a product-aspect score indicates detected evidence but does not guarantee that the evidence volume is large or that the automated label is correct. These distinctions are maintained in the database, results tables, and interpretation.

The FastAPI site exposed four linked modules. Recommend accepted feature priorities and price filters and returned Top-N results with contribution-based explanations. Phones supported product and evidence-coverage inspection. Evidence presented the pipeline walkthrough and downloadable CSV outputs so the reported funnel, aspect labels, and score rows could be audited. Evaluation displayed the star-text sanity check and the accumulated feedback-session metrics. All four modules read from the same frozen SQLite research database.

3.14 Data Analysis and Evaluation

3.14.1 Descriptive Corpus Analysis

Descriptive analysis reports the raw and retained record counts, exclusion reasons, listing coverage, reviews per listing, sentence count, aspect frequencies, sentiment frequencies, product-aspect score coverage, and evidence counts. These statistics establish the evidence base before feature scores or recommendations are interpreted.

The principal corpus values are 4,079 stored reviews across 223 smartphones, 3,388 retained reviews across 222 smartphones, 31,683 sentences, 13,914 aspect-sentiment records, 1,240 score rows, and 221 scored smartphones. Price observations were available for 217 smartphones. Aspect and sentiment distributions are reported in the results chapter because they are outcomes of the implemented pipeline.

3.14.2 ABSA Technical Checks

The supplied implementation does not contain a completed independently gold-labelled benchmark large enough to support a defensible Macro F1 claim. Therefore, precision, recall, Macro F1, confusion matrices, and end-to-end aspect-sentiment pair F1 are not reported as completed results for the primary pipeline.

ABSA is instead examined through the following technical checks:

1. coverage of all six aspect categories and all three sentiment classes;
2. inspection of linked source sentences and prediction provenance;
3. review-star and predicted text-polarity agreement as a broad sanity check; and
4. analysis of low-frequency aspects, neutral classifications, and likely rule failures.

Star-text agreement is not a gold-standard measure because a review may give a high overall rating while criticizing one feature. It is used only to identify major directional inconsistency. A manually labelled stratified sample remains the appropriate future method for calculating aspect and sentiment performance metrics.

3.14.3 Recommendation Evaluation

Recommendation behavior is evaluated on the frozen product pool using predefined weight profiles. The profiles include balanced, battery-focused, camera-focused, display-focused, performance-focused, design-focused, and price-or-value-focused configurations. For each profile, the proposed feature-weighted ranking and Amazon-star baseline are generated under the same available price filters.

The analysis examines whether changing the feature priorities changes the final order and whether the proposed and baseline methods return different products. This demonstrates responsiveness to the intended recommendation logic; it does not by itself prove that one ranked list is objectively superior.

The implemented evaluation interface stored one-to-five phone ratings for displayed recommendations. Twelve valid feedback sessions produced 90 phone ratings. NDCG@3 summarized whether highly rated phones appeared near the top of each recommendation list, while the mean rating and the proportion of four- or five-point ratings summarized acceptance. The current evaluation produced NDCG@3 of approximately 0.56, a mean satisfaction rating of approximately 3.49 out of 5, and approximately 56% high ratings. These outcomes were interpreted as preliminary because the sample was small and no baseline-rated sessions had been completed. The

proposed and Amazon-star outputs should be counterbalanced in a larger comparison before superiority is assessed.

3.14.4 Explanation Evaluation

Explanation evaluation emphasizes faithfulness and coverage. A faithful explanation must identify factors that genuinely contributed to the stored recommendation score. Automated checks compare the stated strengths, weaknesses, scores, weights, contribution values, mention counts, imputation flags, and coverage status with the structured recommendation object.

Explanation coverage is the proportion of recommendations for which the required reason structure is produced successfully. The implemented interface generated contribution-grounded reasons and preserved their source scores and evidence counts. The overall phone ratings provided an initial indication of recommendation acceptance, but they were not a dedicated measure of explanation clarity, usefulness, or trust. Those constructs still require a controlled participant evaluation.

3.14.5 Robustness Analysis

Robustness is examined by varying feature-weight profiles, optional price filters, the minimum-evidence threshold, and the shrinkage strength. The analysis observes changes in product eligibility, Top-N membership, rank order, and evidence coverage. Large changes caused by small parameter adjustments indicate that the recommendation is sensitive to sparse evidence and should be reported cautiously.

Table 3.10: Evaluation measures and their interpretation

Evaluation area	Measure or check	Interpretation
Corpus preparation	Funnel counts and exclusion reasons	Documents the evidence retained at each processing stage.
ABSA coverage	Aspect, sentiment, listing, and mention distributions	Shows whether the pipeline produces usable evidence across the defined taxonomy.
ABSA sanity	Review-star and text-polarity agreement	Detects broad directional inconsistency; it is not aspect-level ground truth.

Evaluation area	Measure or check	Interpretation
Recommendation responsiveness	Ranking changes across fixed priority profiles	Checks whether feature weights have the intended computational effect.
Method comparison	Proposed ranking versus Amazon-star baseline	Shows how personalized feature evidence changes Top-N membership and order.
Optional ranking evaluation	NDCG@3 and Spearman correlation from valid relevance sessions	Measures top-rank relevance and order agreement when sufficient user ratings exist.
Explanation faithfulness	Agreement between stated reasons and computed contributions	Confirms that explanations reflect the actual scoring logic.
Explanation coverage	Percentage of recommendations with complete explanations	Identifies failures to produce the required reason structure.
Robustness	Sensitivity to weights, thresholds, shrinkage, and price filters	Shows whether results depend excessively on one configuration.

3.15 Reliability Validity and Quality Assurance

3.15.1 Reliability

Dataset reliability is supported by preservation of the source CSV, stable product identifiers, a frozen SQLite database, explicit exclusion reasons, and separate raw and processed fields. Method reliability is supported by a versioned aspect lexicon, documented contextual rules, deterministic aggregation equations, stored configuration values, and reproducible command-line operations.

The use of an optional external LLM can reduce determinism because service versions, prompts, and model behavior may change. Engine provenance and lexicon fallback records are therefore required. Results intended to demonstrate the reproducible core methodology should be generated with the fixed lexicon configuration or with a fully documented external model and prompt version.

3.15.2 Internal Validity

Internal validity concerns whether the reported recommendation output is produced by the documented calculation. It is supported by using the same stored feature scores

throughout aggregation, ranking, explanation, and evaluation; normalizing weights consistently; comparing the proposed and baseline methods on the same candidate pool; and checking displayed values against the structured results.

Threats include incorrect product grouping, duplicated listings, language-filter errors, aspect ambiguity, polarity-rule errors, and mean imputation of missing aspects. These risks are not eliminated by reproducible code and must remain visible in the interpretation.

3.15.3 Construct Validity

Construct validity is supported by defining each aspect operationally and using the same definitions throughout the pipeline. Nevertheless, the feature score measures the balance of automated sentiment evidence in the available reviews; it does not directly measure laboratory battery endurance, camera resolution, processor speed, build durability, or market price. The phrase *feature performance* therefore refers to perceived performance in the processed review corpus.

Recommendation fit is operationalized as the weighted sum of these review-derived scores. This makes the meaning of the output transparent, but the score should not be interpreted as a universal measure of product quality or as a guarantee of user satisfaction.

3.15.4 External Validity

Generalizability is limited by the use of one marketplace snapshot, English-language filtering, uneven reviews per listing, ASIN-level product identity, and a six-aspect taxonomy. The findings may not transfer directly to other marketplaces, languages, time periods, product categories, or users whose priorities depend on unmodelled features.

3.15.5 Quality Controls

Table 3.11: Reliability validity and quality controls

Quality area	Implemented control	Expected benefit
Source traceability	Preserve raw CSV, ASINs, source metadata, and database relationships	Supports audit from scores back to review evidence.

Quality area	Implemented control	Expected benefit
Raw-data preservation	Keep original review text separate from normalized text and derived records	Prevents undocumented alteration of source evidence.
Product eligibility	Remove known manual and guidebook patterns before analysis	Reduces contamination by non-phone products.
Language and text quality	Store filtering outcomes and exclusion reasons	Makes the 691 exclusions auditable.
ABSA transparency	Use versioned lexicons, contextual rules, linked sentences, and engine provenance	Makes automated labels inspectable.
Sparse-evidence control	Retain counts, apply Bayesian shrinkage, and report coverage and low-evidence flags	Reduces overconfidence in small samples.
Ranking consistency	Normalize weights and use the same candidate pool for method comparison	Supports fair reconstruction of the calculation.
Explanation faithfulness	Generate reasons from stored feature contributions and validate them numerically	Prevents unsupported recommendation claims.
Configuration control	Record aspect set, engine, shrinkage, minimum mentions, and filters	Supports repeatable analytical runs.

3.16 Ethical Considerations

The study analyzes publicly oriented product reviews obtained through a third-party data source for academic research. Only fields required for product grouping, text analysis, baseline comparison, and reproducibility are used. Reviewer names, profile links, and unnecessary personal identifiers are excluded from analysis outputs.

The raw corpus must be stored securely and used according to applicable data-source licensing, platform conditions, institutional requirements, and research-ethics guidance. Public availability does not remove the responsibility to minimize data collection and avoid republishing identifiable review histories. Results are therefore presented in aggregate, with short evidence examples used only where methodologically necessary and without unnecessary identification of reviewers.

The core corpus analysis did not require questionnaire or interview data. The completed pilot evaluation stored session-level phone ratings without direct personal identifiers. Any expanded controlled study of satisfaction, trust, clarity, or perceived usefulness should use participant information, informed consent, voluntary participation, secure storage, and the institutional ethical procedures applicable before further data collection.

The recommendation system is an academic decision-support artefact. It does not represent commercial endorsement, professional purchasing advice, or a guarantee of product quality. Review evidence may be biased, incomplete, manipulated, outdated, or unevenly distributed, and these limitations should be disclosed when presenting recommendations.

3.17 Tools and Technologies

The tools in Table 3.12 support the implemented data and analytical flow. Interface technologies are not treated as the methodology; they only provide access to the same underlying services and stored outputs.

Table 3.12: Tools and technologies used in the implemented methodology

Methodological component	Tool or technology	Role in the data flow
Programming and orchestration	Python 3.12	Runs ingest, preprocessing, ABSA, aggregation, recommendation, evaluation, and export operations.
Source dataset	Bright Data Amazon smartphone reviews and local CSV	Provides the frozen raw corpus.
Tabular processing	Pandas	Validates, transforms, summarizes, and exports research data.
Database	SQLite	Stores source and derived research records locally.
Data-access layer	SQLAlchemy 2.0	Defines entities, relationships, queries, and transaction handling.
Language filtering	langdetect	Identifies English-language reviews for the main corpus.

Methodological component	Tool or technology	Role in the data flow
Sentence segmentation	pySBD	Detects sentence boundaries in retained review text.
Primary ABSA	Versioned domain aspect and sentiment lexicons with contextual rules	Produces reproducible offline aspect-sentiment records.
Optional ABSA	httpx with an OpenAI-compatible API	Supports structured secondary annotation when configured.
Validation and settings	Pydantic and pydantic-settings	Validates analytical requests and controls configuration.
Research service	FastAPI and Uvicorn	Exposes the shared analytical services without changing the methodology.
Command execution	Typer and Rich	Runs reproducible preparation, analysis, statistics, recommendation, and export commands.
Optional acquisition support	Playwright, BeautifulSoup, lxml, and Tenacity	Supports supplementary collection and controlled retry behaviour.
Version control	Git and dated data or configuration files	Tracks code, lexicon, and experiment-state changes.

3.18 Chapter Summary

This chapter presented the methodology used in the implemented smartphone recommendation research. The study adopts a pragmatic, predominantly quantitative, applied design-and-development approach. It uses a cross-sectional frozen Bright Data Amazon review corpus and follows a traceable pipeline from product screening and review preprocessing through sentence segmentation, lexicon-first ABSA, evidence aggregation, Bayesian-smoothed feature scoring, preference weighting, ranking, explanation, and technical evaluation.

The actual data funnel contains 4,079 stored reviews linked to 223 eligible smartphone listings, 3,388 retained English reviews, 31,683 sentences, 13,914 aspect-sentiment records, and 1,240 product-aspect scores covering 221 listings. Usable reviews were available for 222 smartphones, and prices were available for 217 smartphones. These

figures replace the earlier planned sample of approximately 15 phones, manual large-scale annotation, BERT fine-tuning, and a 70:15:15 model split.

Chapter 4 : Results And Evaluation

4.1 Chapter Introduction

This chapter presents the results produced by the implemented smartphone recommendation pipeline. Each dataset or evaluation output is introduced, presented, and then explained briefly. The chapter reports the final corpus, preprocessing outcome, sentence segmentation, aspect and sentiment distributions, feature-score coverage, recommendation logic verification, and alignment with the approved research objectives and questions. Detailed interpretation, comparison with previous studies, implications, and limitations are reserved for Chapter 5.

4.2 Alignment with the Research Aim, Objectives and Questions

The approved research aim was to develop an explainable feature-based smartphone recommendation system using Aspect-Based Sentiment Analysis (ABSA) of user reviews. The system was intended to analyze textual reviews, extract feature-level sentiments, compute feature performance scores, and generate transparent, requirement-driven recommendations without relying on user history. The results were therefore organized around the five approved objectives and four research questions.

Table 4.1: Alignment of research objectives with reported results

Objective	Approved focus	Result evidence
O1	To collect and prepare a structured dataset of smartphone reviews from Amazon for natural language processing and analysis.	Final corpus and preprocessing funnel in Section 4.3.
O2	To identify predefined smartphone features from review text and classify the associated sentiment as positive, negative or neutral using ABSA.	Aspect and sentiment results in Section 4.4.
O3	To determine each smartphone's feature performance score by calculating the percentage of positive sentiment mentions for each feature.	Final raw, shrunk and confidence scores in Section 4.5.

Objective	Approved focus	Result evidence
O4	To develop a recommendation engine that ranks smartphones using weighted feature scores based on user-defined feature preferences.	Recommendation-engine verification in Section 4.6.
O5	To produce AI explanation messages in plain language that support each suggestion using the actual feature scores.	Contribution-based explanation verification in Section 4.6.

The four approved research questions concerned reliable feature-level sentiment extraction, conversion of feature sentiment into performance measures, generation of explainable recommendations, and comparison with conventional rating-based ranking. The database outputs directly supported the first three questions. The fourth was examined through the implemented Amazon-star baseline, star-text sanity checking, and the initial user sessions; however, the absence of baseline-rated runs prevented a completed paired superiority comparison.

4.3 Dataset Preparation Results

The dataset snapshot shows that each review was linked to a smartphone through its Amazon Standard Identification Number (ASIN) and retained product information such as smartphone name, brand, price, site rating, review title, and review text. The full dataset was processed as an offline and reproducible corpus. Manuals and guide books were removed before review-level preprocessing.

Table 4.2: Final dataset and processing funnel

Processing stage	Measured output	Result
Raw reviews stored	Reviews / smartphones	4,079 / 223
Product data coverage	Smartphones with usable reviews / price observations	222 / 217
Spam flag status	Reviews marked	25
Preprocessing exclusions	Reviews excluded for other reasons	691
After preprocessing	Usable English reviews	3,388
Sentence segmentation	Sentences	31,683

Processing stage	Measured output	Result
ABSA output	Aspect-sentiment records	13,914
Feature-score output	Phone-aspect score records	1,240
Final score coverage	Smartphones with at least one score	221

The stored dataset contained 4,079 reviews across 223 smartphones. Preprocessing retained 3,388 usable English reviews, equivalent to 83.1%, and excluded 691 records. Twenty-five reviews were marked by the spam heuristic, with those flags overlapping the exclusion decisions. The retained reviews produced 31,683 sentences, or approximately 9.4 sentences per retained review. At least one usable review remained for 222 smartphones, and the number of retained reviews per phone ranged from 1 to 24, with an average of approximately 15.3. The 223 smartphones comprised Apple 175, Google 16, Motorola 14, Samsung 12, and OnePlus 6; price observations were available for 217 smartphones.

Table 4.3: Reasons for excluding reviews during preprocessing

Exclusion reason	Reviews	Share of exclusions
Non-English language	663	95.95%
All-capital text	12	1.74%
No alphabetic content	10	1.45%
Too short	3	0.43%
Generic-only content	3	0.43%
Total	691	100.00%

Language screening accounted for 663 of the 691 exclusions and was therefore the main cause of review removal. The remaining exclusions were caused by low-information or unusable text. Twenty-five reviews were also identified by the spam flag; where a flagged review met an exclusion rule, it was counted under the applicable exclusion category rather than added again to the total.

4.4 Aspect and Sentiment Results

The ABSA stage generated 13,914 records across the six predefined smartphone aspects. A record represented an identified aspect together with its positive, neutral or

negative sentiment label. Table 4.4 and Figure 4.1 show the resulting aspect distribution.

Table 4.4: Distribution of aspect-sentiment mentions

Aspect	Mentions	Share
Battery	4,479	32.19%
Display	2,450	17.61%
Price	2,313	16.62%
Camera	2,019	14.51%
Performance	1,403	10.08%
Design	1,250	8.98%
Total	13,914	100.00%

Battery was the most frequently identified aspect, producing 4,479 mentions and representing 32.19% of all aspect-sentiment records. Display and price followed with 2,450 and 2,313 mentions respectively. Design recorded the fewest mentions, although its 1,250 records still provided a substantial evidence base at corpus level.

The overall sentiment output contained 6,736 positive, 5,744 neutral, and 1,434 negative records. Positive sentiment represented 48.41% of the output, neutral sentiment represented 41.28%, and negative sentiment represented 10.31%. The complete distribution by aspect is presented in Table 4.5.

Table 4.5: Sentiment distribution for each smartphone aspect

Aspect	Positive	Neutral	Negative	Total
Battery	1,789	2,233	457	4,479
Display	955	1,258	237	2,450
Price	1,562	550	201	2,313
Camera	898	928	193	2,019
Performance	835	323	245	1,403
Design	697	452	101	1,250
Total	6,736	5,744	1,434	13,914

Price produced the highest positive proportion at 67.53%, followed by performance at 59.52% and design at 55.76%. Display had the largest neutral proportion at 51.35%, while performance had the largest negative proportion at 17.46%. These values describe the database output only; the reasons for the observed patterns are examined in Chapter 5.

4.5 Feature-Score Results

Aspect-sentiment records were aggregated by smartphone and aspect. For every phone-aspect pair, the database stored positive, neutral and negative counts, total mentions, a raw score, a Bayesian-shrunk feature score, and an evidence-confidence value. This produced 1,240 phone-aspect score records across 221 smartphones.

Table 4.6: Fields in the final phone-aspect score dataset

Column	Meaning
smartphone_id	Internal smartphone/listing identifier
phone_name	Smartphone display name joined from the phone table
aspect	Battery, camera, display, performance, design or price
positive_count	Number of positive mentions (P)
neutral_count	Number of neutral mentions (U)
negative_count	Number of negative mentions (N)
mention_count	Total evidence: $P + U + N$
raw_score	Unshrunk sentiment score from 0 to 1
feature_score_shrunk	Bayesian-shrunk score used for ranking
confidence	Evidence-strength value based on the mention count

$$\text{Raw Score} = (P + 0.5 \times U) / (P + U + N)$$

The raw score awarded full credit to positive evidence, half credit to neutral evidence, and no credit to negative evidence. Bayesian shrinkage with $k = 5$ then adjusted sparse phone-aspect scores towards the corpus mean before ranking. Confidence was calculated as mentions / (mentions + 3), so a score supported by more mentions received a stronger evidence value.

Of the 223 smartphones, 221 had at least one feature score, giving product-level scoring coverage of 99.10%. The 1,240 observed phone-aspect records represented 92.68% of the 1,338 theoretically possible combinations of 223 phones and six aspects. Across the 221 scored smartphones, the database contained an average of approximately 5.61 scored aspects per phone. Design evidence was available for 200 smartphones, equivalent to 89.69% of the product set.

The final implementation therefore retained the approved goal of deriving feature performance from sentiment evidence, while improving the original percentage-positive calculation through neutral weighting, evidence confidence, and Bayesian shrinkage. The shrunk score, rather than an unstable score based on very few mentions, was used by the recommendation engine.

4.6 Recommendation and Explanation Evaluation

The recommendation engine accepted user importance values for battery, camera, display, performance, design, and price. The entered values were normalized and multiplied by the corresponding shrunk feature scores. Smartphones were then ordered by the resulting weighted score. Missing aspect values were imputed using the relevant candidate-set mean and identified as imputed rather than being assigned a score of zero. A coverage value showed how much of the requested preference weight was supported by observed review evidence.

The explanation component used the same weighted contributions that produced the ranking. Consequently, a recommendation reason was linked to the user's most important aspects and the phone's strongest supported feature scores. This preserved traceability between user input, feature evidence, final score, ranking, and explanation. The dataset snapshot also showed mixed evidence within individual reviews, such as positive price or camera comments appearing together with battery or performance concerns, supporting the need for aspect-level rather than overall-text analysis.

Table 4.7: Verification of recommendation and explanation functions

Component	Evaluation check	Result
Preference input	Six aspect importance values accepted	Implemented
Weight processing	Entered values normalized before scoring	Verified by formula

Component	Evaluation check	Result
Ranking	Shrunk feature scores combined using normalized weights	Implemented
Sparse evidence	Missing values imputed and coverage reported	Implemented
Baseline	Amazon site rating retained for comparison	Available
Explanation	Reasons derived from weighted feature contributions	Implemented
Pilot user evaluation	12 sessions; 90 phone ratings; mean satisfaction; NDCG@3	3.49/5 mean; 56% high ratings; NDCG@3 = 0.56
Star-text sanity check	Directional polarity agreement for 2,925 reviews	Approximately 67% agreement
Paired baseline evaluation	Amazon-star lists rated under the same protocol	Incomplete (0 baseline-rated runs)

The Evaluation tab contained results from 12 feedback sessions and 90 phone ratings. Mean satisfaction was approximately 3.49 out of 5, and about 56% of the ratings were four or five. NDCG@3 was approximately 0.56, indicating weak-to-moderate Top-N ordering because highly rated phones were not consistently placed at the first positions. Separately, predicted review-text polarity agreed directionally with Amazon star ratings for approximately 67% of 2,925 comparable reviews. Agreement was stronger for four- and five-star reviews (approximately 70–79%) and weaker for one- to three-star reviews (approximately 33–37%). Star-text agreement was treated as a sanity check rather than an accuracy measure because an overall star rating can coexist with positive and negative opinions about different features. No Macro-F1 or confusion matrix was reported because a gold-labelled benchmark was not available. The Amazon-star ranking baseline was implemented, but zero baseline-rated runs had been completed, so comparative superiority was not claimed.

4.7 Evaluation against the Research Questions

Table 4.8: Results aligned with the approved research questions

Approved research question	Result demonstrated by the implementation
RQ1: How can sentiment at the feature level be reliably extracted from reviews of smartphones?	Preprocessing, sentence segmentation and ABSA produced 13,914 records across six aspects.
RQ2: How can sentiment scores for features be combined to create useful performance metrics?	Counts were aggregated into raw scores, shrunk ranking scores and evidence-confidence values for 1,240 phone-aspect combinations.
RQ3: How can explainable recommendations be produced using feature scores?	Normalized user weights were combined with shrunk feature scores, and explanations were generated from the strongest weighted contributions.
RQ4: When compared to conventional rating-based systems, does feature-based recommendation enhance interpretability?	The Amazon-star baseline was implemented. Pilot results showed mean satisfaction of approximately 3.49 and NDCG@3 of approximately 0.56, but no baseline-rated runs were available; comparative superiority was therefore not established.

The first three questions were addressed by the database and implemented processing flow. For the fourth question, the initial user results showed that the preference-weighted approach could produce acceptable recommendations for some users, but mean satisfaction and NDCG@3 remained mixed. Because no baseline-rated runs were available, the evidence did not establish that the proposed ranking was more relevant or interpretable than Amazon-star ordering.

4.8 Chapter Summary

This chapter presented the results and evaluation of the developed ABSA-based smartphone recommendation system. From 4,079 stored reviews across 223 smartphones, 3,388 usable English reviews were retained. Sentence segmentation produced 31,683 sentences, and ABSA generated 13,914 aspect-sentiment records. Battery was the most frequently discussed aspect, while the overall output contained 6,736 positive, 5,744 neutral, and 1,434 negative mentions. Aggregation produced 1,240 phone-aspect score records and at least one feature score for 221 smartphones.

The implemented scoring method used neutral weighting, Bayesian shrinkage, and confidence values before user-weighted ranking and contribution-based explanation. Star-text agreement was approximately 67% for 2,925 reviews. Twelve sessions and 90 phone ratings produced mean satisfaction of approximately 3.49 out of 5 and NDCG@3 of approximately 0.56. These initial results supported feasibility but not superiority because the paired baseline evaluation remained incomplete. Chapter 5 discusses the meaning and limitations of these findings.

Chapter 5 : Discussion

5.1 Chapter Introduction

This chapter discusses the results presented in Chapter 4 in relation to the research problem, objectives, questions, and literature reviewed in Chapter 2. The discussion focuses on the quality and coverage of the review corpus, the observed aspect and sentiment patterns, the revised feature-scoring method, personalized ranking, and contribution-based explanations. It also considers the value of the implemented approach, the extent to which the research questions were answered, and the limitations that affect interpretation of the findings.

5.2 Review Corpus and Data Quality

The final dataset was substantially larger and broader than the small product sample originally anticipated. The database contained 4,079 reviews linked to 223 smartphone listings. Preprocessing retained 3,388 usable English reviews, representing 83.1% of the stored corpus. This demonstrates that the source data contained a sufficient quantity of usable review text for feature-level analysis, while also confirming that direct use of unfiltered marketplace data would have introduced considerable noise.

Non-English text accounted for 663 of the 691 excluded reviews. This result supports the decision to apply language screening because the implemented ABSA resources were designed for English. At the same time, removing these reviews restricts the findings to English-language customer expression and may underrepresent the experience of multilingual users. Previous multilingual ABSA research similarly identifies language imbalance, weak supervision, and domain transfer as important challenges (Satria et al., 2025; Ratniasih et al., 2025; Di Nuovo et al., 2024). The exclusion therefore improved internal consistency but reduced linguistic coverage.

The removal of all-capital, non-alphabetic, very short, and generic-only records further reduced low-information input. The spam flag identified 25 reviews, showing that review credibility remained relevant even after basic cleaning. This is consistent with the observation that promotional, repeated, or misleading reviews can distort recommendation results and user trust (Jin et al., 2026). However, an automated spam flag cannot establish authenticity with certainty. It should be interpreted as a data-

quality control rather than proof that every retained review represents a verified and unbiased customer experience.

The retained reviews produced 31,683 sentences, averaging approximately 9.4 sentences per review. Sentence segmentation was important because smartphone reviews frequently contain several feature opinions in one long narrative. Separating the text into smaller units reduced the risk of assigning the sentiment of one feature to another. This supports the use of aspect-level processing instead of treating each review as a single positive or negative document, which is a central advantage of ABSA in multi-feature product analysis (Soni et al., 2024; Singhal et al., 2025; Kumari et al., 2025).

5.3 Aspect and Sentiment Patterns

Battery produced 4,479 mentions and represented 32.19% of all aspect-sentiment records, making it the most frequently discussed feature. Display, price, and camera followed, while performance and design received fewer mentions. The dominance of battery suggests that users commonly describe practical experiences that are difficult to judge from technical specifications alone, including endurance, charging behavior, and battery health. This reinforces the value of review text as a source of real-use evidence rather than relying only on manufacturer specifications or overall ratings.

The lower number of design mentions does not necessarily indicate that design is unimportant. It indicates only that design was discussed less frequently in the collected reviews or was expressed through terms outside the fixed vocabulary. Similarly, performance may have been described indirectly through heating, lag, gaming, responsiveness, or application behavior. A predefined aspect scheme improves consistency and interpretability, but implicit and uncommon expressions may remain undetected. This reflects the wider trade-off between transparent aspect rules and the contextual flexibility of transformer-based ABSA models (Mansilla et al., 2025; Aziz et al., 2024; Jayakody et al., 2024).

Positive sentiment formed 48.41% of the 13,914 aspect records, neutral sentiment formed 41.28%, and negative sentiment formed 10.31%. The large neutral proportion is meaningful because many reviews contain factual statements, balanced descriptions, or weakly evaluative language. Giving neutral evidence partial value in the score avoided treating such mentions as either fully positive or fully negative. Nevertheless, the distribution also shows class imbalance. A system that produces relatively few

negative labels may fail to capture subtle criticism, particularly when sentiment is expressed through sarcasm, contrast, or context-dependent language.

Price had the highest positive proportion, while performance had the highest negative proportion. These patterns should not be interpreted as universal judgments about smartphone prices or performance. They describe the composition of this particular Amazon corpus, which included products from different brands, price levels, conditions, and review volumes. The findings are therefore useful for explaining the database used by the recommender, but they do not provide market-wide estimates of customer opinion.

5.4 Feature Scoring and Evidence Coverage

The final implementation refined the original percentage-positive feature score. Positive mentions received full value, neutral mentions received half value, and negative mentions received no value in the raw score. Bayesian shrinkage with $k = 5$ then adjusted scores supported by few mentions towards the corpus mean. This change did not alter the purpose of Research Objective 3, which was to convert feature sentiment into comparable performance measures. Instead, it reduced the possibility that a phone with one positive mention would rank above a phone supported by many consistently favorable mentions.

The confidence calculation provided a separate measure of evidence strength. This distinction between score and confidence is important. A high score indicates favorable sentiment, while high confidence indicates that the score is supported by a larger number of observations. Combining these ideas makes the recommendation more auditable because the system can distinguish a strong feature result from a sparsely supported result rather than presenting both with equal certainty.

At least one feature score was available for 221 of the 223 smartphones, giving product-level coverage of 99.10%. The database contained 1,240 phone-aspect score records, equivalent to 92.68% of all possible phone-aspect combinations, with an average of 5.61 scored aspects per scored phone. This level of coverage indicates that the six-aspect scheme was suitable for most products in the corpus. However, coverage was not complete, and design evidence was available for 200 phones. Mean imputation allowed incomplete products to remain eligible, but the accompanying coverage value was necessary so that estimated evidence was not mistaken for observed review sentiment.

This evidence-aware scoring approach is consistent with the literature that treats aspect sentiment as structured recommendation information rather than an end result of classification (Musto et al., 2017; Liu et al., 2021; Haghighi and Moradian Zadeh, 2025). The main contribution is not a new statistical estimator in isolation, but the integration of review evidence, shrinkage, confidence, user weighting, and explanations within one traceable decision process.

5.5 Personalized Recommendation and Cold Start

The recommendation engine used explicit user importance values for battery, camera, display, performance, design, and price. These values were normalized and combined with the shrunk feature scores. Consequently, the same smartphone database could produce different rankings for users with different priorities. This is appropriate for smartphone selection because a phone that is attractive to a camera-focused user may not be the strongest choice for a user who values battery endurance or price.

The explicit weighting method differs from collaborative recommendation, which commonly learns preferences from previous ratings, clicks, or purchases. By asking the user to state priorities directly, the system reduces the new-user cold-start problem and does not require long-term behavioral tracking. Review and content signals have previously been used to reduce dependence on sparse interaction histories (Volkovs et al., 2017; Lin, 2022; Danish et al., 2026). The present study extends that principle through a simple and visible weighting mechanism.

This simplicity is also a limitation. User-entered importance values may be inconsistent, and they do not capture complex interactions between features. For example, camera quality may depend partly on performance, display quality may influence perceived design, and price may change the acceptability of weaknesses in other areas. More complex recommender models can learn such interactions, but they may reduce transparency. The implemented weighted-sum model therefore prioritized understandable and reproducible decisions over hidden preference modelling.

The Amazon star-rating baseline was implemented for comparison. The initial evaluation comprised 12 sessions and 90 phone ratings, with mean satisfaction of approximately 3.49 out of 5 and NDCG@3 of approximately 0.56. About 56% of the phone ratings were four or five. These results show that the preference-weighted ranking was acceptable to some users but that the ordering of preferred phones remained weak to moderate. Because the evaluation contained zero baseline-rated runs, it could

not quantify whether the feature-based ranking was more relevant than star-based ordering.

5.6 Explainability of the Recommendations

The explanation module used the same weighted contributions that generated the recommendation score. This design is important because an explanation should describe the actual basis of the ranking rather than attach a generic statement after the decision. Feature-based explanations can make trade-offs clearer by identifying the aspects that contributed most strongly and by showing when evidence was missing or imputed (Dietz et al., 2022; Mansilla et al., 2025; Danish et al., 2026).

The approach supports user-centered explainability. Model-centered interpretation explains internal prediction behavior, whereas user-centered explanation answers why a particular phone suits the buyer's declared priorities. By linking the recommendation to feature scores, weights, mention counts, and coverage, the system provides a reason that can be checked against the calculation. This responds to the literature calling for explanations that reveal feature trade-offs and improve the transparency of high-involvement product decisions (Chen et al., 2026; Zhu, 2025; Gajula, 2025).

However, structural faithfulness does not by itself prove that users find an explanation clear or useful. The generated statement may accurately reflect the ranking while still being too technical, too long, or insufficiently persuasive. The mean satisfaction score of approximately 3.49 indicated mixed acceptance of the recommendations as a whole, but it was not a dedicated explanation-quality measure. The study can therefore claim that explanations were contribution-grounded, while clarity, usefulness, and trust require a targeted evaluation.

5.7 Discussion of the Research Questions

Table 5.1: Discussion of findings against the research questions

Research question	Discussion
RQ1	Feature-level sentiment was extracted through cleaning, segmentation, fixed aspect identification and three-class sentiment assignment. The 13,914 resulting records demonstrate operational coverage, but classification accuracy remains unquantified without gold labels.

Research question	Discussion
RQ2	Sentiment evidence was transformed into raw scores, Bayesian-shrunk scores and confidence values. This produced stable and comparable phone-aspect measures while retaining evidence counts.
RQ3	Normalized user priorities were combined with shrunk scores, and explanations were generated from the largest weighted contributions. This created a traceable path from review evidence to recommendation reason.
RQ4	The star-rating baseline and proposed method were implemented. Twelve sessions and 90 phone ratings produced mixed satisfaction (approximately 3.49/5) and weak-to-moderate NDCG@3 (approximately 0.56). Because no baseline-rated runs were completed, comparative superiority was not established.

Overall, the implementation provided direct practical answers to the first three research questions and partial empirical evidence for the fourth. The strongest evidence concerned corpus processing, aspect-sentiment coverage, score generation, traceability, and the feasibility of preference-based ranking. The star-text sanity check and pilot user metrics strengthened the evaluation, but the small sample and absence of paired baseline ratings prevented a claim of comparative superiority.

5.8 Research and Practical Implications

The research demonstrates that ABSA outputs can be used beyond sentiment reporting. The pipeline converts unstructured review text into feature evidence, aggregates that evidence into product profiles, combines the profiles with explicit preferences, and explains the resulting ranking. This addresses the integration gap identified in Chapter 2, where smartphone ABSA studies often stop at classification and recommender studies often rely on ratings or latent interaction patterns (Soni et al., 2024; Musto et al., 2017; Liu et al., 2021).

For smartphone buyers, the approach provides a more detailed basis for comparing products than one overall star rating. For retailers, the aspect distributions can indicate which features dominate customer feedback. Manufacturers may use the same aggregated evidence to identify frequently praised or criticised areas. These uses should remain evidence-aware: a feature score should be presented together with mention volume, confidence, and coverage rather than as an absolute measure of product quality.

The system also provides a practical example of explainable decision support without behavioral profiling. Because users state their priorities directly and the explanation refers to visible contributions, the recommendation process is easier to inspect. This is valuable where transparency and user control are more important than maximizing predictive complexity.

5.9 Limitations and Threats to Validity

Several limitations affect the interpretation of the findings. First, the corpus was an offline snapshot of Amazon review data. Product availability, prices, ratings, and customer opinions change over time, so the results represent the collected period rather than a permanent market condition. Review counts were also uneven across smartphones, which may give heavily reviewed products more stable feature profiles than sparsely reviewed products.

Second, the final reproducible ABSA engine was lexicon-first. This made the processing transparent and executable without a labelled training dataset, but it may miss implicit aspects, uncommon wording, sarcasm, long-distance dependencies, and context-sensitive sentiment. Transformer research reports stronger contextual performance in mobile and technical reviews (Soni et al., 2024; Singhal et al., 2025; Aziz et al., 2024). Without a manually labelled test set, the accuracy, precision, recall, Macro-F1, and confusion matrix of the implemented annotations could not be established.

Third, the analysis was limited to six predefined aspects and English-language reviews. Other factors such as software support, network reception, speakers, charging speed, repairability, and after-sales service were outside the primary model. A recommendation may therefore omit an attribute that is important to a particular buyer. Fourth, missing aspect scores were imputed using the candidate-set mean. This was preferable to assigning zero, which would incorrectly treat missing evidence as poor performance. Nevertheless, imputation reduces product differentiation and introduces estimated rather than directly observed evidence. The coverage value partly controls this risk, but users must still be informed when a recommendation depends on imputed features.

Finally, the user evaluation was preliminary. Twelve sessions and 90 phone ratings yielded mean satisfaction of approximately 3.49 and NDCG@3 of approximately 0.56, but the sample was small and contained no baseline-rated runs. Star–text agreement of

approximately 67% was directional and did not replace a gold-labelled ABSA test. Explanation clarity, usefulness, and trust were not measured separately. Accordingly, the findings support feasibility, coverage, traceability, and functional implementation; they do not establish validated classification accuracy or statistical superiority over conventional recommendation systems.

5.10 Chapter Summary

The findings indicate that an offline Amazon review corpus can provide substantial feature-level evidence for explainable smartphone recommendation. Battery dominated the discussion, sentiment composition differed across aspects, and high phone-aspect coverage supported multi-feature scoring for most products. Neutral weighting, Bayesian shrinkage, and confidence values improved the treatment of sparse evidence, while explicit user weights supported transparent personalisation without historical interaction data. Contribution-based explanations connected the final rank to its evidence. Star–text agreement was approximately 67%, while 12 feedback sessions produced mean satisfaction of approximately 3.49 and NDCG@3 of approximately 0.56. The approach therefore achieved the integration objective and produced promising initial user evidence, although gold-label validation and paired baseline evaluation remain necessary.

Chapter 6 : Conclusions And Recommendations

6.1 Chapter Introduction

This chapter presents the conclusions drawn from the development and evaluation of the explainable feature-based smartphone recommendation system. The conclusions are based on the implemented data pipeline and the results reported in Chapters 3 and 4. The study shows that customer-review text can be transformed into traceable feature-level evidence and combined with user priorities to produce personalised smartphone rankings and plain-language reasons. The chapter also identifies practical recommendations for responsible use and for strengthening the evidence produced by the system.

6.2 Conclusions of the Study

The research aim was to develop an explainable feature-based smartphone recommendation system using Aspect-Based Sentiment Analysis of user reviews. This aim was achieved at the level of system development and functional implementation. The completed process begins with a frozen Amazon review dataset, removes ineligible products and unusable review text, segments the retained reviews, detects six smartphone aspects, assigns positive, neutral, or negative sentiment, aggregates the evidence into feature scores, applies user-defined preferences, ranks eligible smartphones, and generates explanations from the same contributions used in the ranking calculation.

The final analysis dataset contained 223 smartphones and 3,388 usable English reviews retained from 4,079 stored reviews. These reviews produced 31,683 sentences and 13,914 aspect-sentiment annotations. Feature scoring generated 1,240 smartphone-aspect rows, and 221 smartphones had at least one available feature score. These figures demonstrate that the pipeline retained sufficient evidence to construct multi-feature profiles for most products in the dataset while also exposing the small number of products and aspects for which review evidence was unavailable.

The results further show that review evidence was not distributed equally across features. Battery was the most frequently discussed aspect, with 4,479 mentions, followed by display, price, camera, performance, and design. Across all aspects, 6,736

mentions were positive, 5,744 were neutral, and 1,434 were negative. This uneven distribution supports the use of feature-level analysis: an overall star rating cannot show which product characteristics attracted the praise, uncertainty, or criticism expressed by reviewers.

The implemented score also improved on a simple percentage-positive measure. Positive mentions received full value, neutral mentions received half value, and negative mentions received no value in the raw feature score. Bayesian shrinkage with $k = 5$ reduced the influence of extreme scores based on very few mentions, while a separate confidence value represented evidence strength. As a result, recommendation quality was not represented by sentiment alone; the implementation also retained the amount and coverage of the supporting evidence.

The recommendation component demonstrated that explicit user priorities can be applied without purchase history or previous interaction data. The six preference values are normalised and combined with the Bayesian-shrunk feature scores. This makes each final score reconstructable and provides a practical response to the new-user cold-start problem. Missing aspect values are handled through candidate-set mean imputation, but the imputation and evidence coverage remain visible so that absence of review evidence is not silently presented as certain product performance.

Explainability was achieved through contribution-based messages. The stated strengths, weaknesses, and trade-offs are derived from the stored feature scores, user weights, evidence counts, and weighted contributions. The system can therefore explain why a product appears in a recommendation list instead of presenting an unsupported ranking. This establishes computational faithfulness, although it does not by itself prove that every user will consider an explanation clear, useful, or trustworthy.

Taken together, the first three research questions were answered by the implemented and traceable flow: feature-level sentiment was extracted, sentiment evidence was converted into usable feature metrics, and those metrics were used to generate explainable recommendations. The fourth question concerned comparison with conventional rating-based recommendation. The Amazon-star baseline and comparison mechanism were implemented, and the pilot evaluation produced mean satisfaction of approximately 3.49 out of 5 and NDCG@3 of approximately 0.56 from 12 sessions and 90 phone ratings. However, zero baseline-rated runs were available. The study therefore concludes that the approach is feasible and transparent, with mixed initial ranking results, but it does not claim statistically demonstrated superiority over the baseline.

6.3 Recommendations

- 1. Present scores together with evidence.** Recommendations should display feature scores with mention counts, confidence, and coverage. A high score from limited evidence should not be presented with the same certainty as one supported by many mentions.
- 2. Use the system as decision support.** The ranking should assist comparison rather than represent an absolute judgement. Buyers should also consider current price, warranty, availability, specifications, and requirements not covered by reviews.
- 3. Maintain a controlled and current evidence base.** Product, review, price, and rating data should be refreshed at defined intervals. Dataset dates, configuration, lexicon version, and engine provenance should be retained for every run.
- 4. Validate automated annotations before making accuracy claims.** Human annotators should label a stratified sample covering all aspects and sentiments. Agreed labels should then be used to calculate precision, recall, F1, and confusion matrices.
- 5. Complete the baseline and user-centred evaluation.** The feature-based and Amazon-rating methods should use the same candidates and counterbalanced presentation. Relevance, ranking agreement, clarity, usefulness, and trust require adequate valid sessions.
- 6. Preserve transparency when improving the model.** Any contextual or hybrid ABSA method should retain sentence links, version information, confidence, and a reproducible fallback so rankings and explanations remain traceable to their evidence.

6.4 Chapter Summary

The study achieved its main development aim by integrating review preprocessing, lexicon-first ABSA, evidence-aware feature scoring, user preference weighting, ranking, and contribution-based explanation within one auditable flow. The strongest conclusions concern feasibility, coverage, traceability, and functional personalisation. The pilot evaluation added initial evidence of mixed user satisfaction and weak-to-moderate ranking quality, but classification accuracy, explanation-specific user outcomes, and paired comparative quality require further empirical evaluation. The next chapter states the limitations and future work.

Chapter 7 Limitations and Future Work

7.1 Chapter Introduction

This chapter identifies the boundaries of the completed research and the work needed to extend it. The limitations do not invalidate the implemented pipeline; instead, they define how far the reported results can be generalized and which claims remain untested. Future work is therefore linked directly to the data, language processing, scoring, recommendation, and evaluation limitations observed in the study.

7.2 Limitations of the Study

7.2.1 Dataset Scope and Representativeness

The analysis used one frozen, cross-sectional Amazon review export. It represents the products, reviews, prices, and customer opinions available in that snapshot and cannot describe later market changes. The dataset was also limited to 223 eligible smartphone listings, and retained reviews were unevenly distributed across products, ranging from 1 to 24 per phone. Scores for heavily discussed phones are therefore generally supported by more evidence than scores for sparsely reviewed phones. Amazon customers may also differ from buyers on other platforms or in other regions, which limits external validity.

Review data are observational and may contain selection bias, duplicated opinions, promotional content, or manipulated reviews. The implemented quality rules and spam indicators reduced obvious problems but could not verify the authenticity of every reviewer or statement. Product listings and ASINs may also combine variants whose storage, color, region, or configuration affects the user experience.

7.2.2 Language and ABSA Limitations

The main corpus was restricted to English because the aspect and sentiment resources were designed for English-language processing. This improved consistency but excluded 663 non-English reviews and prevents the findings from representing multilingual customer expression. Automatic language identification may also misclassify short or mixed-language text.

The reproducible ABSA engine was lexicon-first. Its versioned terms and contextual rules make decisions inspectable, but they may miss implicit aspects, uncommon

vocabulary, sarcasm, comparative statements, long-distance dependencies, and opinions whose meaning depends on wider context. Neutral labels can also represent genuine mixed or uncertain opinions as well as cases where no sufficiently strong cue was found. Because no manually gold-labelled test set was completed, the study cannot report reliable accuracy, precision, recall, Macro-F1, or a confusion matrix for the annotations.

7.2.3 Feature Scoring and Recommendation Limitations

The analysis used a closed taxonomy of battery, camera, display, performance, design, and price. These features are important but do not cover every purchase factor, such as durability, software support, connectivity, security, repairability, storage, warranty, or after-sales service. The feature scores also summarize reviewer sentiment rather than objective laboratory performance or manufacturer specifications.

Bayesian shrinkage and confidence values reduce overinterpretation of small samples but cannot remove bias in the underlying reviews. Mean imputation allows a product with missing aspect evidence to remain eligible, yet it reduces differentiation and may conceal an important evidence gap if coverage indicators are ignored. The weighted-sum ranking is transparent, but it assumes that feature contributions are additive and independent. It does not learn interactions, changing preferences, or differences between stated importance and actual choice behavior.

7.2.4 Evaluation and Generalization Limitations

The study verified the end-to-end data flow and the numerical basis of the ranking and explanations. The user evaluation comprised only 12 sessions and 90 phone ratings. Mean satisfaction was approximately 3.49 out of 5 and NDCG@3 was approximately 0.56, while about 56% of the ratings were four or five. These values provide preliminary evidence but are too limited for broad generalisation. Spearman correlation and dedicated measures of explanation clarity, usefulness, and trust were not reported. Most importantly, zero baseline-rated runs were available, so the current evidence does not establish that the proposed ranking is more relevant or interpretable than Amazon-star ordering.

7.3 Future Work

- 1. Create a gold-labelled ABSA benchmark.** Multiple trained annotators should label a balanced sentence sample. The benchmark can quantify the current engine and support fair comparison with transformer, large-language-model, and hybrid approaches.
- 2. Extend multilingual and implicit-aspect coverage.** Future versions should evaluate major source languages and code-mixed text. Hybrid methods should also be tested for implicit features, sarcasm, comparisons, and mixed sentiment while retaining provenance.
- 3. Broaden and refresh the evidence base.** Longitudinal data from multiple marketplaces and regions would improve representativeness. Helpfulness, verified-purchase status, recency, duplication, and spam probability could become evidence-quality weights.
- 4. Improve uncertainty and missing-evidence treatment.** Alternative Bayesian or hierarchical models should be compared with current shrinkage. Confidence intervals, recency weighting, and stricter handling of missing high-priority aspects should also be tested.
- 5. Expand preference and ranking models.** Additional aspects, price constraints, must-have conditions, and feature interactions could be supported. Advanced models should be compared with the transparent weighted-sum method.
- 6. Conduct a controlled comparative user study.** Participants should assess counterbalanced feature-based and Amazon-rating lists. Relevance, task time, clarity, usefulness, trust, and satisfaction should follow a predefined plan and adequate sample size.
- 7. Evaluate explanations as user-facing evidence.** Short contribution summaries, trade-off statements, and selected review evidence should be compared. Faithfulness checks should remain mandatory, while participants assess clarity and practical usefulness.

7.4 Chapter Summary

The principal limitations arise from the single-platform snapshot, English-only processing, uneven review coverage, a fixed six-aspect taxonomy, lexicon-first ABSA, simplified preference aggregation, the small pilot evaluation, and the absence of gold-labelled and paired baseline results. Future work should prioritise validated annotations

and a larger controlled comparison with Amazon-star ordering. Multilingual processing, broader and more current evidence, stronger uncertainty modelling, richer preferences, and dedicated explanation evaluation can then extend the system while preserving its traceability.

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